CSSMedia User Guide



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Features

CSSMedia offers these main features.

- **XML format agnostic.** CSSMedia works with any well-formed XML. It does not assume anything about element names or attributes.
- **Any grammar.** XML is parsed with TypesXML, which handles any vocabulary. You can pass the grammars of your documents to CSSMedia with an OASIS Catalog.
- **CSS based.** The stylesheet is the only source of styling. Selector matching, the cascade, inheritance and value resolution follow the CSS specifications.
- **PDF for paged media.** Page sizes, margins, page breaks, named pages, headers, footers and page numbers are controlled with CSS paged media features.
- **Multilingual support.** Documents can be in any language and script, including right-to-left languages and CJK languages.
- **Hyphenation by language.** Text is hyphenated with the patterns of the language declared with `xml:lang`.
- **Fonts.** OpenType, TrueType, font collections, WOFF and WOFF 2.0 fonts, loaded from `@font-face` rules or a fonts folder, plus the 14 standard PDF fonts.
- **Shared fonts folder.** You can pass a central fonts folder to CSSMedia and reuse it from any project.
- **Generated content.** Counters, quotes, list markers, named strings, running elements, cross-references and tables of contents.
- **Links and bookmarks.** Internal and external links and PDF bookmarks.
- **Images.** PNG, JPEG, JPEG 2000 and WebP images.
- **SVG and MathML.** SVG and MathML found in a document are drawn as images.

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Get started

CSSMedia turns an XML file and a CSS file into a PDF.

To produce a PDF, install CSSMedia, prepare your XML and run it from the command line with your stylesheet. These topics show how:

- Install CSSMedia
- Prepare your XML
- Run CSSMedia
- Your first stylesheet
- Try the samples

Install CSSMedia

Follow these steps to build CSSMedia from source or to install it as a global command.

Procedure

1. Clone the CSSMedia project and build it with npm:

   ```
   git clone https://github.com/maxprograms-com/CSSMedia.git
   cd CSSMedia
   npm install
   npm run build
   ```

2. Check that the build works by running CSSMedia:

   ```
   node dist/cssmedia.js
   ```

   Without parameters, CSSMedia displays its help.

3. To use the `cssmedia` command from any folder, install the package globally:

   ```
   npm i -g file:cssmedia-x.x.x.tgz
   ```

   Download or build the package first, and replace `x.x.x` with its version.

Prepare your XML

CSSMedia generates a PDF from almost any well-formed XML file, provided that you give it an appropriate CSS file.

Some XML projects consist of multiple files that need pre-processing before they are ready for publishing. Projects based on DITA or DocBook need to merge all the files that make up a project into one XML file.

- If you work with DITA, use TypesDitac to process your topics and generate one combined XML file for CSSMedia.
- If you work with DocBook, TypesBook helps you merge your files into a single document ready for CSSMedia.

Run CSSMedia

Run CSSMedia from the command line with an XML file, a CSS file and the name of the PDF to write.

This command generates a PDF from your XML and your stylesheet:

```
cssmedia -x document.xml -c document.css -p document.pdf
```

If you built CSSMedia from source and did not install it globally, run `node dist/cssmedia.js` instead of `cssmedia`.

Running CSSMedia without parameters displays this help:

```
Usage: cssmedia -x <file> -c <file> -p <file> [-f <fontsFolder>] [-g <catalog>]

  -x, --xml <file>       XML file to format
  -c, --css <file>       CSS stylesheet to apply
  -p, --pdf <file>       PDF file to write
  -f, --fonts <folder>   (optional) alternative fonts folder
  -g, --catalog <file>   (optional) OASIS catalog to use when parsing XML
```

Use `-f` to pass a fonts folder that you reuse from any project. Use `-g` to pass an OASIS Catalog when your XML refers to grammars.

Your first stylesheet

CSSMedia takes two files: an XML document and a CSS stylesheet. XML contains the document's elements and text. CSS contains rules that describe how selected elements should be styled.

Procedure

1. Look at the XML

   Suppose your XML contains this element:

   ```
   <chapter>
       <title>Using the device</title>
       <p>Follow these steps before starting.</p>
   </chapter>
   ```

   The names `chapter`, `title`, and `p` belong to this XML document. They are not special CSS names. You can use the names from your own XML in a stylesheet.

2. Write a rule

   A CSS rule has a selector and a declaration block:

   ```
   title {
       font-size: 18pt;
   }
   ```

   The selector `title` chooses XML elements with that name. The declaration `font-size: 18pt;` sets a property to a value. The braces contain the declarations for the selected elements.

   You can add more declarations to the same rule:

```
title {
    font-family: serif;
    font-size: 18pt;
    color: navy;
}
```

Each declaration has a property name, a colon, a value, and a semicolon. CSSMedia checks property values against the CSS syntax it supports. A selector only matches when it describes elements in your XML.

3. Use names from your own XML

   The element names in the examples are placeholders. If your XML uses `heading` instead of `title`, write `heading` as the selector:

   ```
   heading {
       font-size: 18pt;
   }
   ```

   CSSMedia does not infer that `heading` is a title or that `p` is a paragraph. You decide which XML names to select and which CSS properties to set.

4. Generate the PDF

   Save the XML as `document.xml` and the stylesheet as `document.css`, then run CSSMedia:

   ```
   cssmedia -x document.xml -c document.css -p document.pdf
   ```

   Open `document.pdf`. The `title` element is styled by the rule you wrote.

Try the samples

The CSSMedia repository includes sample documents, stylesheets, fonts and the PDFs they produce. Follow these steps to generate a PDF from a sample and change its stylesheet.

About this task

The `samples` folder of the repository has these samples:

Sample	Content
xml_gap	A DocBook article in English, Arabic, Japanese and Russian, formatted with one stylesheet, `xml_gap.css`.
Swordfish	The User Guide of a real product, written in DITA and merged into one XML file with TypesDitac and with the DITA Open Toolkit. The stylesheets `ditac.css`, `ditac_hyphens.css` and `dita-ot.css` format the merged files.
fonts	The Noto fonts that the stylesheets load with `@font-face` rules.

Each sample comes with the PDF that CSSMedia generated from it.

Procedure

1. Open a terminal in the `samples` folder.

2. Generate the PDF of the English article:

   ```
   cssmedia -x xml_gap/xml_gap.xml -c xml_gap.css -p xml_gap/xml_gap.pdf
   ```

   The XML files declare a DocBook DTD. Without a catalog, CSSMedia prints the warning `External subset not found: docbook/docbookx.dtd` and still generates the PDF. To avoid the warning, pass a catalog that resolves the DTD with `-g`, such as the one in TypesBook.

3. Open the PDF and compare it with the one in the repository.
4. Edit `xml_gap.css` and run the command again.

   For example, change the `font-size` of `article` or the `color` of `section > title`, and see what changes in the PDF.

5. Generate the Swordfish User Guide:

   ```
   cssmedia -x Swordfish/typesditac/swordfish.xml -c ditac.css -p Swordfish/
   typesditac/swordfish.pdf
   ```

Results

The `xml_gap.css` stylesheet uses these techniques:

Result	CSS
Local fonts	`@font-face` rules with `src: url("./fonts/NotoSans-Regular.ttf")`
Page size, margins and a footer with the document title and the page number	`@page` with `@bottom-left` and `@bottom-right`, `string-set: doc-title content()` on the title, and `counter(page)`
A different font for each language	Selectors such as `article[lang|="ar"]` and `article[lang|="ja"]`
Hyphenation in English and Russian	`hyphens: auto` on `article:lang(en)` and `article:lang(ru)`
Right-to-left text in Arabic	`direction: rtl`
Stricter line breaking in Japanese	`line-break: strict`
Links from an XML attribute	`-cssmedia-link: attr(url)` on `ulink`
An image from an XML attribute	`content: attr(fileref url)` on `imagedata`
The photo at the start side of the text, on the right in Arabic and on the left in the other languages	`float: inline-start` on `mediaobject`
An arrow after each link	`ulink::after { content: " \2197"; }`

Result	CSS
Code listings in a monospace font	`font-family: "Noto Sans Mono", monospace` with `white-space: pre`

Selectors

A selector tells CSSMedia which element or elements in the XML file a CSS rule applies to. A rule has a selector followed by braces. Inside the braces, write declarations that set property values:

```
chapter {
    color: navy;
}
```

Here, `chapter` is the selector and `color: navy;` is a declaration. CSSMedia reads XML names and attributes as data. It does not assign vocabulary-specific meanings to them.

The guides are grouped by selector type and matching behavior:

- Selector reference: a compact reference to the selector forms and their PDF behavior.
- Basic selectors and namespaces: element names, IDs, classes, attributes and XML namespaces.
- Combinators: selecting elements by their relationship to other elements.
- Pseudo-classes: structural tests, selector logic, language and other states.
- Pseudo-elements: selecting generated content, text parts and other special targets.
- Specificity: how selectors help resolve competing declarations.

Basic selectors and namespaces

Basic selectors match an element by its name, attributes or namespace. Combine selector parts when you want to require more than one condition.

Element names and any element

Selectors 4 §5.1, Type (tag name) selector; Selectors 4 §5.2, Universal selector.

A type selector is an element name, such as `chapter` or `title`. It matches elements with that name. The universal selector `*` matches any element.

```
chapter { color: navy; }
* { font-family: serif; }
```

An unprefixed name uses the default namespace declared with `@namespace`. Without a declared default namespace, it matches an element with no namespace.

IDs and classes

Selectors 4 §6.6, Class selectors; Selectors 4 §6.7, ID selectors.

An ID identifies an element. Write `#` followed by the ID value to select it. CSSMedia checks the `id` and `xml:id` attributes.

```
#overview { color: navy; }
```

A class selector starts with a period, as in `.warning`. CSSMedia accepts this syntax, but does not use it to test an XML attribute. Selectors Level 4 leaves the meaning of class membership to the document language, and XML does not define one universal class attribute.

If your XML uses an unnamespaced `class` attribute containing space-separated class names, test it with an attribute selector. The ~= operator checks for one whole, space-separated value:

```
[class~="warning"] { color: red; }
```

This matches an element whose `class` attribute contains `warning` as one of its values. It does not make `.warning` a working class selector in CSSMedia.

Attributes

Selectors 4 §6, Attribute selectors.

An attribute selector checks an XML attribute. Put its name in square brackets to require that the attribute exists. Add an operator and value to compare its contents.

```
[role] { color: navy; }
[role="notice"] { color: navy; }
[class~="warning"] { color: red; }
[lang|="en"] { font-family: serif; }
[href^="https"] { color: blue; }
[href$=".pdf"] { color: purple; }
[data-kind*="reference"] { color: teal; }
[role="notice" i] { color: navy; }
```

= compares the whole value. ~= checks for a space-separated word. |= matches the value itself or a value followed by a hyphen. ^=, $= and *= check for text at the start, end or anywhere in the value. The `i` flag ignores ASCII letter case. The `s` flag requests case-sensitive comparison.

An attribute selector without a namespace prefix checks an attribute with no namespace. Use a declared prefix to select an attribute in a namespace.

XML namespaces

CSS Namespaces 3 §3, Declaring namespaces: the @namespace rule.

XML elements and attributes can belong to namespaces. A CSS namespace declaration assigns a prefix to a namespace URI. Use that prefix in a selector to choose elements in that namespace.

```
@namespace svg url("http://www.w3.org/2000/svg");

svg|rect { color: navy; }
*|rect { color: teal; }
|rect { color: gray; }
```

`svg|rect` selects `rect` elements in the declared SVG namespace. `*|rect` selects `rect` elements in any namespace. `|rect` selects `rect` elements with no namespace. The CSS prefix is an alias for the URI. It does not need to match the prefix used in the XML.

A bare attribute selector such as `[id]` always selects an attribute with no namespace. An undeclared prefix makes that selector invalid. CSSMedia ignores a style rule if any selector in its selector list is invalid.

Combinators

A combinator describes the relationship between two elements. Put the related element's selector on the left and the element you want to match on the right.

Descendant and sibling relationships

Selectors 4 §14, Combinators.

Use a space to select a descendant at any depth. Use > for a direct child, + for the next element sibling and ~ for any later element sibling with the same parent.

```
chapter title { color: navy; }
chapter > title { color: teal; }
title + paragraph { color: gray; }
title ~ paragraph { color: purple; }
```

The first rule selects every `title` inside a `chapter`, even when other elements are between them. The second selects only titles that are direct children of a chapter. The last two select paragraphs based on their position after a title.

Column combinator

Selectors 5 §8.1, Column combinator (||).

The column combinator `||` relates elements that belong to the same table column.

```
col || td { color: navy; }
```

Reference combinators

Selectors 5 §9.1, Reference combinators /ref/.

A reference combinator relates elements through an attribute that refers to an element ID.

```
section /aria-labelledby/ title { color: navy; }
```

The name between slashes is the reference attribute. CSSMedia accepts this syntax and uses the referenced element relationship when matching.

Pseudo-classes

A pseudo-class starts with : and tests something about an element, such as its position in the XML tree, its language, or an interaction state. Some pseudo-classes can be evaluated from the XML tree. Others depend on interaction states, a media timeline, or meanings defined by a document vocabulary, which CSSMedia does not use when processing a PDF.

CSSMedia accepts selector syntax only when it recognizes that syntax. A recognized pseudo-class does not necessarily match an element. The guides below distinguish pseudo-classes CSSMedia can use to match XML elements from valid pseudo-classes it accepts but does not match in PDF processing.

Child and sibling position

"Selectors 4 §13, Tree-Structural pseudo-classes".

Structural pseudo-classes select elements by their position among siblings in the XML tree. They do not depend on how the elements are displayed.

```
chapter:first-child { color: navy; }
chapter:last-child { color: teal; }
chapter:only-child { color: gray; }
chapter:first-of-type { color: purple; }
```

`:first-child`, `:last-child` and `:only-child` count all element siblings. The `-of-type` forms count siblings with the same element name and namespace. Text, comments and processing instructions are not counted as element siblings.

Nth-position selectors

"Selectors 4 §13.3, Child-indexed Pseudo-classes".

Use an `:nth-*` pseudo-class to select an element by its numbered position among siblings. Positions start at one. `odd` and `even` select alternating positions; `An+B` describes a repeating sequence.

```
chapter:nth-child(odd) { color: navy; }
chapter:nth-child(2n + 1) { color: teal; }
chapter:nth-child(2 of [status="active"]) { color: purple; }
```

The `of` clause in `:nth-child()` and `:nth-last-child()` filters the siblings before CSSMedia counts them. The `-last-` forms count from the end. The `-of-type` forms count only siblings with the same element name and namespace.

CSSMedia matches `:nth-child()`, `:nth-last-child()`, `:nth-of-type()` and `:nth-last-of-type()`.

Column position

"Selectors 5 §8.2, :nth-col() pseudo-class"; "Selectors 5 §8.3, :nth-last-col() pseudo-class".

`:nth-col()` and `:nth-last-col()` select cells by their position in a table column.

Root, empty and scope

"Selectors 4 §13.1, :root pseudo-class"; "Selectors 4 §13.2, :empty pseudo-class"; "Selectors 4 §8.4, The Reference Element Pseudo-class: :scope".

`:root` selects the root element of the XML document. `:empty` selects an element with no child elements or non-whitespace text. Whitespace-only text, comments and processing instructions do not prevent an element from matching `:empty`.

`:scope` selects the current scoping element. When CSSMedia matches a selector without another scope, the document root is the scope.

```
:root { color: navy; }
chapter:empty { display: none; }
:scope > chapter { color: teal; }
```

CSSMedia matches these pseudo-classes using the XML tree. The nesting selector `&` is different from a pseudo-class. It refers to the parent selector inside a nested rule.

Selector-list and relational pseudo-classes

"Selectors 4 §4.2, The Matches-Any Pseudo-class: :is()"; "Selectors 4 §4.3, The Negation (Matches-None) Pseudo-class: :not()"; "Selectors 4 §4.4, The Specificity-adjustment Pseudo-class: :where()"; "Selectors 4 §4.5, The Relational Pseudo-class: :has()".

These pseudo-classes let you combine selectors. `:is()` matches an element if it matches at least one selector in its argument. `:where()` works the same way but adds no specificity. `:not()` excludes elements that match its argument. `:has()` selects an element based on a related element.

```
chapter:not([status="draft"]) { color: navy; }
:is(chapter, appendix) > title { color: teal; }
:where([compact]) paragraph { font-size: 10pt; }
section:has(> figure) { color: purple; }
title:has(+ paragraph) { color: gray; }
```

`:not()` makes the selector invalid if its argument contains an invalid selector. `:is()` and `:where()` discard invalid selectors from their lists. CSSMedia also accepts `:matches()` as an alias for `:is()`. A `:has()` argument cannot contain another `:has()` or a pseudo-element.

Language and direction

"Selectors 4 §7.1, The Directionality Pseudo-class: :dir()"; "Selectors 4 §7.2, The Language Pseudo-class: :lang()".

Use `:lang()` to select elements by language. CSSMedia checks the language of the element, which is the nearest `xml:lang` value on the element or its ancestors, or the language that the `-cssmedia-lang` property gives. See Document language.

```
:lang(fr) { font-family: serif; }
```

If no language is set, `:lang("")` matches. For `:dir(ltr)` and `:dir(rtl)`, CSSMedia scans the element's descendant text for the first strong character using the Unicode Bidirectional Algorithm. If there is none, it checks the language of the element, normalized, and uses its bidi classification. If there is no language or it is not classified as bidi, the direction defaults to left-to-right. The CSS `direction` property does not affect `:dir()`.

Links and document location

"Selectors 4 §8.1, The Hyperlink Pseudo-class: :any-link"; "Selectors 4 §8.2, The Link History Pseudo-classes: :link and :visited"; "Selectors 4 §8.3, The Target Pseudo-class: :target"; "Selectors 5 §2.1, The Local Link Pseudo-class :local-link".

CSSMedia uses its `link` property to identify elements with hyperlink destinations. `:link` and `:any-link` select those elements. `:local-link` selects a link to the same document. Its optional integer limits the number of path segments compared.

```
:any-link { color: blue; }
:local-link { color: navy; }
```

CSSMedia does not track browsing history, so it does not distinguish visited links. `:visited` and `:target` are accepted but do not match elements. The `link` property is a CSSMedia extension; it supplies link destinations for XML elements.

User interaction

"Selectors 4 §9, User Action Pseudo-classes"; "Selectors 5 §3.1, The Interest Pseudo-classes: :interest-source and :interest-target".

User-action pseudo-classes describe states such as pointing at an element, activating it, or giving it keyboard focus. Interest pseudo-classes describe a user indicating interest in an element. These states require interaction; they do not occur during PDF processing.

`:hover` means a pointing device is over an element. `:active` means the element is being activated. `:focus` means it has input focus; `:focus-visible` means that focus should be indicated; and `:focus-within` also matches an element containing the focused element. `:interest-source` and `:interest-target` identify the source and target of a user's declared interest.

CSSMedia accepts these selectors, but none match elements in PDF processing.

Media playback and time

"Selectors 4 §10.1, Media Playback State: the :playing, :paused, and :seeking pseudo-classes"; "Selectors 4 §10.2, Media Loading State: the :buffering and :stalled pseudo-classes"; "Selectors 4 §10.3, Sound State: the :muted and :volume-locked pseudo-classes"; "Selectors 5 §5.1, The Current-element Pseudo-class: :current"; "Selectors 5 §5.2, The Past-element Pseudo-class: :past"; "Selectors 5 §5.3, The Future-element Pseudo-class: :future".

`:playing`, `:paused` and `:seeking` describe playback state. `:buffering` and `:stalled` describe media loading. `:muted` and `:volume-locked` describe sound state. The time-dimensional pseudo-classes `:current`, `:current(...)`, `:past` and `:future` describe elements in relation to a timeline, such as the current playback position or speech rendering.

CSSMedia accepts these pseudo-classes, but none match elements in a PDF. CSSMedia has no media playback state or timeline to evaluate.

Input and interface state

"Selectors 4 §12.1.1, The :enabled and :disabled Pseudo-classes"; "Selectors 4 §12.1.2, The Mutability Pseudo-classes: :read-only and :read-write"; "Selectors 4 §12.1.3, The Placeholder-shown Pseudo-class: :placeholder-shown"; "Selectors 4 §12.1.4, The Automatic Input Pseudo-class: :autofill"; "Selectors 4 §12.1.5, The Default-option Pseudo-class: :default"; "Selectors 4 §12.2.1, The Selected-option Pseudo-classes: :checked, :unchecked, and :indeterminate"; "Selectors 4 §12.3.1, The Validity Pseudo-classes: :valid and :invalid"; "Selectors 4 §12.3.2, The Range Pseudo-classes: :in-range and :out-of-range"; "Selectors 4 §12.3.3, The Optionality Pseudo-classes: :required and :optional"; "Selectors 4 §12.3.4, The User-interaction Pseudo-classes: :user-valid and :user-invalid"; "Selectors 5 §4.1.1, The Empty-Value Pseudo-class: :blank".

These pseudo-classes describe form controls and interface states. `:enabled` and `:disabled` test whether a control can be used. `:read-write` tests whether its value can be changed; `:placeholder-shown` tests whether placeholder text is visible; `:autofill` tests whether a value was filled automatically; and `:default` identifies a default control. `:checked`, `:unchecked` and `:indeterminate` test selection state. `:valid` and `:invalid` test constraint validation; `:in-range` and `:out-of-range` test a value against its allowed range. `:required` and `:optional` test whether a value is required. `:user-valid` and `:user-invalid` test validity after user interaction. `:blank` tests whether a control's value is empty.

Their meaning depends on behavior defined by a host language, such as HTML. CSSMedia treats XML names and attributes as data and does not infer that an element is a form control.

CSSMedia accepts `:enabled`, `:disabled`, `:read-write`, `:placeholder-shown`, `:autofill`, `:default`, `:checked`, `:unchecked`, `:indeterminate`, `:blank`, `:valid`, `:invalid`, `:in-range`, `:out-of-range`, `:required`, `:optional`, `:user-valid` and `:user-invalid`, but none match elements. `:read-only` matches every element.

Document-language-defined states

"Selectors 5 §6, Exposing custom state: the :state() pseudo-class"; "Selectors 5 §7, Heading Structures: the heading pseudo-classes :heading, and :heading()".

Some pseudo-classes depend on states or semantics supplied by a document vocabulary. `:state(name)` checks a custom element state. `:heading` and `:heading()` depend on which elements the vocabulary defines as headings and what levels they represent.

CSSMedia accepts `:state()`, `:heading` and `:heading()`, but they do not match elements because CSSMedia does not assign custom states or heading semantics to XML elements.

Interface and presentation states

"Selectors 4 §11.1, Collapse State: the :open pseudo-class"; "Selectors 4 §11.2, Popover State: the :popover-open pseudo-class"; "Selectors 4 §11.3, Modal (Exclusive Interaction) State: the :modal pseudo-class"; "Selectors 4 §11.4, Fullscreen Presentation State: the :fullscreen pseudo-class"; "Selectors 4 §11.5, Picture-in-Picture Presentation State: the :picture-in-picture pseudo-class".

`:open` identifies an open or expanded element. `:popover-open` identifies an open popover. `:modal` identifies an element in modal presentation. `:fullscreen` identifies an element shown in fullscreen mode, and `:picture-in-picture` identifies an element presented in picture-in-picture mode.

These are interface or presentation states that CSSMedia does not provide. It accepts the pseudo-classes, but they do not match elements.

Defined elements

"Selectors 4 §5.4, The Defined Pseudo-class: :defined".

`:defined` tests whether the document vocabulary distinguishes fully defined elements from undefined ones. XML as processed by CSSMedia has no separate defined or constructed state, so `:defined` matches every element.

Pseudo-elements

A pseudo-element starts with `::` and targets a special part or state associated with an element. It appears after the element selector:

```
paragraph::first-letter { color: navy; }
paragraph::before { content: "Note: "; }
```

Text parts

CSS Pseudo-Elements 4 §2, Typographic Pseudo-elements.

`::first-line` selects the first formatted line of text. `::first-letter` selects the first typographic letter or punctuation group.

```
paragraph::first-line { font-weight: bold; }
paragraph::first-letter { font-size: 150%; }
```

Use these selectors to style the first formatted line or typographic letter of an element.

Generated content, markers and footnotes

CSS Pseudo-Elements 4 §4, Tree-Abiding Pseudo-elements; CSS GCPM 3 §1.1.1, The string-set property; CSS GCPM 3 §1.1.1.1, The content() function; CSS GCPM 3 §1.1.2, The string() function; CSS GCPM 3 §2.6, The footnote-call pseudo-element; CSS GCPM 3 §2.7, The footnote-marker pseudo-element.

`::before` and `::after` identify generated-content positions before or after an element's content. The `content` property supplies their content. `::marker` targets a list marker. The footnote pseudo-elements identify a footnote call and marker.

```
paragraph::before { content: "Note: "; }
item::marker { color: navy; }
```

Paged CSS also defines `string-set` to copy an element's text into a named string for reuse, for example in a page margin box. See Named strings for how to use `string-set` and `string()`.

The `content` property specifies what `::before` and `::after` contain. A selector alone does not add text. The list marker's content can come from `content` or the list-marker properties. Footnote pseudo-elements mark the call and marker parts of a footnote.

Highlight pseudo-elements

CSS Pseudo-Elements 4 §3.1, Selecting Highlighted Content.

`::selection` identifies text selected by the user. `::search-text` identifies text found by a search. `::target-text` identifies text targeted by a document link. `::spelling-error` and `::grammar-error` identify ranges marked by spelling and grammar checking. `::highlight(name)` selects a named text range registered through the Custom Highlight API.

These ranges are defined by selection, search, navigation, spelling and grammar checking, or script-provided highlights. They are not ordinary XML elements.

Element-backed pseudo-elements

CSS Pseudo-Elements 4 §4, Tree-Abiding Pseudo-elements; CSS Pseudo-Elements 4 §5, Element-backed Pseudo-Elements; CSS Position 4 §3.2, The ::backdrop Pseudo-Element.

`::placeholder` targets placeholder text in a control. `::file-selector-button` targets a file-upload control's button. `::details-content` targets the expandable content of a disclosure control. `::backdrop` targets the backdrop behind an element presented in the top layer.

These pseudo-elements refer to parts of controls or elements defined by the document language. Their meaning depends on the host language's control and presentation behavior.

CSSMedia also accepts `::before::marker`, `::after::marker`, `::first-letter::prefix` and `::first-letter::suffix`. These select marker content associated with

generated `::before` or `::after` content, and punctuation around a `::first-letter` part. A supported state pseudo-class can follow a pseudo-element, but interaction states such as `:hover` do not match.

Specificity

Specificity helps the cascade choose between declarations from matching rules. It is one part of the cascade, after origin, importance and layer priority.

Comparing selector specificity

Selectors 4 §15, Calculating a selector's specificity.

Selectors contribute specificity according to their parts. An ID selector contributes more than an attribute selector or pseudo-class. Those contribute more than an element name or pseudo-element.

```
chapter { color: gray; }
[role="notice"] { color: teal; }
#important-notice { color: navy; }
```

If all other cascade priorities are equal, the ID rule wins over the attribute rule, and the attribute rule wins over the element-name rule. If competing rules have the same specificity, source order decides and the later declaration wins.

`:where()` contributes no specificity. For `:is()`, `:not()` and `:has()`, specificity is based on the most specific selector in the argument. The `&` nesting selector contributes specificity according to the parent selector list.

CSSMedia accepts class-selector syntax such as `.warning`, but does not use it to test an XML attribute. When your XML stores space-separated class names in an unnamespaced `class` attribute, use `[class~="warning"]` instead. This is an attribute selector, so its specificity is that of an attribute selector. See Basic selectors and namespaces.

Define pages

Use an @page rule to define the page box used for paged output. The page context contains ordinary CSS property declarations and the @page descriptors `size`, `bleed`, `marks` and `page-orientation`. Ordinary properties set page-box values. Descriptors set page-specific dimensions, marks and orientation.

About this task

CSS Paged Media 3 §4.1, The @page Rule; §4.3, @page rule grammar.

Procedure

1. @page rule syntax

   Write @page, an optional comma-separated list of page selectors, and a block:

   ```
   @page {
       size: A4;
   }

   @page :first, :left {
       margin-top: 25mm;
   }
   ```

   The first rule has no selector and applies to every page. The second applies when either selector matches. A selector consists of an optional page-type name followed by zero or more page pseudo-classes. Do not put whitespace between the name and pseudo-classes or between pseudo-classes. Page-type names are case-sensitive identifiers; `auto` is never a matching page-type name.

   Page selector list and grammar; Page-type selectors.

   Each declaration block can contain page-box properties and the four descriptors described below. The rule can also contain page-margin at-rules, described in Page-margin rules.

2. Set page size with `size`

   The `size` descriptor sets the target page-box dimensions and orientation. Its initial value is `auto`. It accepts `auto`, one or two lengths, or a page-size name optionally combined with `portrait` or `landscape`.

   CSS Paged Media 3 §7.1, Page size.

```
@page {
    size: A4;
}

@page landscape {
    size: A4 landscape;
}

@page square {
    size: 150mm;
}

@page custom {
    size: 210mm 297mm;
}
```

`size` values

- `auto` lets the user agent choose the page-box size and orientation to suit its output. CSSMedia uses A4 as its default page size.
- `portrait` makes the shorter sides horizontal. Without a page-size name, it orients the user agent's chosen page size this way.
- `landscape` makes the longer sides horizontal. Without a page-size name, it orients the user agent's chosen page size this way.
- A single length, such as `210mm`, sets both width and height, producing a square page box.
- Two lengths set the width first and the height second. A length must not be negative.

The standard `<page-size>` names correspond to these sheet dimensions:

Value	Sheet dimensions
`A3`	297 mm wide by 420 mm high
`A4`	210 mm wide by 297 mm high
`A5`	148 mm wide by 210 mm high
`B4`	250 mm wide by 353 mm high
`B5`	176 mm wide by 250 mm high
`JIS-B4`	257 mm wide by 364 mm high
`JIS-B5`	182 mm wide by 257 mm high
`letter`	8.5 inches wide by 11 inches high
`legal`	8.5 inches wide by 14 inches high
`ledger`	11 inches wide by 17 inches high

- `portrait` and `landscape` can each be combined with one page-size name, as in `size: A4 landscape`.

- If the declaration appears in a media query that tests page size or orientation, CSS ignores it. This avoids using the selected page size to decide which page size to select.
- CSS Paged Media 3 §7.1, Page size.

11. Set page margins

    Margins set the space between the page box and the page area. The page area is where the document's boxes are laid out. CSSMedia uses the page margins when it calculates that area. If you do not set page margins, CSSMedia's user-agent page rule sets each margin to `1in`.

    CSS Paged Media 3 §4.1, The @page Rule; CSS Box Model 4 §3.1, Physical margin properties; §3.2, Margin shorthand.

    `margin`

    The `margin` shorthand sets all four page margins. It accepts one to four lengths such as `12pt`, percentages such as `5%`, or `auto`. A negative length is allowed. Its initial value is `0`, and it is not inherited.

    - One value sets all four sides.
    - Two values set the top and bottom, then the right and left.
    - Three values set the top, left and right, then the bottom.
    - Four values set the top, right, bottom and left.

    Percentages are resolved against the page size in the corresponding axis. For example, left and right margins use the page width; top and bottom margins use the page height.

    ```
    @page {
        size: A4;
        margin: 20mm 25mm;
    }
    ```

    Individual margin properties

    Each longhand sets the margin on one physical side of the page box. Each accepts a length such as `12pt`, a percentage such as `5%`, or `auto`; negative lengths are allowed. Each has an initial value of `0` and is not inherited.

    - `margin-top` sets the top margin.
    - `margin-right` sets the right margin.
    - `margin-bottom` sets the bottom margin.
    - `margin-left` sets the left margin.

    For example, use wider inner margins for two-sided pages:

    ```
    @page :left {
        margin-left: 30mm;
    }

    @page :right {
        margin-right: 30mm;
    }
    ```

20. Select pages

A page selector matches pages by page type or page position. You can combine a page type name with page pseudo-classes.

CSS Paged Media 3 §4.2, Page selectors; CSS GCPM 3 §3.1, Page selectors.

Page-type names

A page-type name in an `@page` selector matches pages assigned that name by the `page` property. The name is a case-sensitive custom identifier. For example, `@page wide` selects pages whose type is `wide`.

CSS Paged Media 3 §4.2, Page selectors; §8.1, Using named pages.

Page pseudo-classes

- `:first` matches the first printed page.
- `:left` matches a left page, as determined by the document's page progression.
- `:right` matches a right page, as determined by the document's page progression.
- `:blank` matches content-empty pages produced by forced page breaks.
- `:nth(An+B)` matches pages by their document position. `A` and `B` define an arithmetic sequence: `:nth(2n)` selects pages 2, 4, 6 and so on, and `:nth(3n+1)` selects pages 1, 4, 7 and so on.

The optional `of` clause in `:nth(An+B of page-group)` counts within the named page group instead of from the start of the document. A page group is created when an element has both a named page value and a forced page break. Each instance starts a new group.

CSS GCPM 3 §3.2, Page groups.

```
@page :first {
    margin-top: 30mm;
}

@page :nth(2n) {
    margin-top: 25mm;
}
```

26. Assign a named page with `page`

The `page` property assigns a page type to a box. Its initial value is `auto`; it is not inherited. It accepts `auto` or a custom identifier that names an `@page` rule.

CSS Paged Media 3 §8.1, Using named pages.

```
@page wide {
    size: A4 landscape;
}

wide-table {
    page: wide;
}
```

When `page` is `auto`, the used page type comes from the nearest ancestor whose value is not `auto`; on the root element it is the unnamed page type. A named page type requires content to start on a page of that type, inserting a forced page break when necessary.

27. Set print marks with `marks`

    The `marks` descriptor requests marks outside the page box. Its initial value is none. Marks help trim or align printed sheets and are visible only when the page box is smaller than the printable area.

    CSS Paged Media 3 §7.2, Crop and registration marks.

    - `none` requests neither crop nor registration marks.
    - `crop` requests short lines that indicate the page-box edges for trimming.
    - `cross` requests cross-shaped registration marks used to align sheets.
    - `crop cross` requests both types. The grammar also accepts them in the opposite order.

```
@page {
    marks: crop cross;
}
```

32. Set the bleed area with `bleed`

    The `bleed` descriptor sets how far the bleed area extends beyond the page box before rendering is clipped. Its initial value is `auto`.

    CSS Paged Media 3 §7.3, Bleed area.

    - `auto` computes to `6pt` when `marks` includes `crop`, and to zero otherwise.
    - A length, such as `3mm`, sets the outward extension on every side. The value may be negative.

```
@page {
    marks: crop;
    bleed: 3mm;
}
```

35. Rotate the laid-out page with `page-orientation`

    The `page-orientation` descriptor changes the orientation of the laid-out page in the output, after layout. It does not change the page-box dimensions used during layout. Its initial value is `upright`; rotation applies when the output medium supports it.

    CSS Paged Media 3 §7.1.2, Rotating the printed page.

    - `upright` leaves the page orientation unchanged.
    - `rotate-left` rotates the page a quarter turn counter-clockwise after layout.
    - `rotate-right` rotates the page a quarter turn clockwise after layout.

    Use `size: A4 landscape` to lay out a landscape page box. Use `page-orientation` when the page should be laid out at one orientation and then rotated in the output.

39. Page-margin rules

    Page-margin at-rules go inside an `@page` rule. Each one identifies a box in a page margin. The `content` property determines whether the box is generated: `none` or `normal` generates no box; a string generates text. A margin box can also use a counter, such as `counter(page)`, to show the page number.

CSS Paged Media 3 §5, Page-margin boxes; §5.1, At-rules for page-margin boxes; §5.2, Populating page-margin boxes; CSS Content 3, `content` ; CSS Paged Media 3 §6.1, Page-based counters.

The page reference lists all 16 page-margin at-rules and describes where each box sits. Its declarations section lists the properties you can use and links to the specification's property list.

For example, put a running title in the top-center box and the page number in the bottom-center box:

```
@page {
    @top-center {
        content: "Document title";
    }

    @bottom-center {
        content: counter(page);
    }
}
```

See Named strings for using element content in page-margin boxes.

Page orientation, bleed and marks

Named strings

Named strings let a stylesheet save content from elements as a value and retrieve that value elsewhere. They are intended for paged content such as running headers and footers.

About this task

Before writing a rule, inspect your XML and find the element whose text or attribute you want to reuse. A CSS rule has a selector followed by declarations inside braces. The selector chooses which XML elements the declarations apply to. In these examples, a name such as `title` is a type selector: it matches elements with that name. It has no special meaning to CSSMedia. Replace example names such as `title`, `group`, and `item` with names that actually occur in your XML. Each declaration sets a property using `property: value;` syntax. See Selectors overview for other ways to choose elements.

Procedure

1. Assign named strings with `string-set`

   CSS Content 3 §2.7.1, The string-set property; CSS GCPM 3 §1.1.1.1, The content() function.

   CSS Content 3 specifies none or one or more comma-separated pairs, each containing a `<custom-ident>` followed by one or more strings. CSSMedia's parser also accepts a `<content-list>` after each name. That broader form is a CSSMedia parser extension, not the specified value grammar. The initial value is none , which assigns no named strings.

   The examples below use the parser extension so you can see its accepted syntax. CSSMedia currently validates `string-set` but does not save named strings or resolve the `string()` function.

   ```
   h1 {
       string-set: chapter-title content(text);
   }
   ```

   Here, `h1` is the selector, `string-set` is the property, and the text after the colon is its value. `chapter-title` is a name you choose for the saved value. It does not need to occur in the XML. `content(text)` takes the text from each selected element. Use the actual element name from your XML instead of `h1` if it is different. A later matching element can replace the saved value with its own text. One declaration can save multiple values:

   ```
   h1 {
       string-set: chapter-title content(text), chapter-label "Chapter "
   counter(chapter);
   }
   ```

   The <custom-ident> is an author-defined identifier. It cannot be a CSS-wide keyword or none.

   The content is one or more items. The `<content-list>` grammar allows these forms:

   - A `<string>` contributes its written text.
   - `counter(<counter-name>, <counter-style>?)` contributes the formatted value of the innermost counter with that name on the element.

- `counters(<counter-name>, <string>, <counter-style>?)` contributes the formatted values of every in-scope counter with that name, from outermost to innermost, joining them with the required string.
- `content()` reads text from the element or one of its generated parts.
- `attr(<attr-name> <syntax>?, <fallback>?)` contributes an attribute value. The optional syntax controls how that value is parsed; the fallback is used when the attribute is missing or cannot be parsed.

Counter functions

CSS Lists 3 §4.3, Nested Counters and Scope; CSS Lists 3 §4.7, Outputting Counters; CSS Counter Styles 3, Counter styles.

Counters are values maintained by CSS. Use `counter-reset`, `counter-increment`, or `counter-set` to create or change them. A counter name is a `<custom-ident>` and cannot be none or a CSS-wide keyword. To include a counter value in generated content, use `counter()` or `counters()`.

A reset of a counter with the same name inside an existing counter's scope creates a nested counter. This lets repeated structures use one counter name at every depth. On an element inside those scopes:

- `counter(name)` returns only the innermost counter named `name`.
- `counters(name, separator)` returns all in-scope counters named `name`, ordered outermost first and innermost last. `separator` is a CSS string inserted between the formatted values.

In the sample XML below, a `group` can contain `item` elements and another nested `group`. These names are examples. If your XML uses different element names, change the selectors to match them. The `group` rule starts an `item` counter at zero for each group. The `item` rule increases that counter by one for each item. The `item::before` rule adds generated text before each item's own text. Its `content` value calls both counter functions: `counter(item)` shows only the current, innermost number; `counters(item, ".")` shows the number at each nesting level, separated by a period.

```
<group>
    <item>First</item>
    <item>
        Second
        <group>
            <item>Nested first</item>
            <item>Nested second</item>
        </group>
    </item>
</group>
```

```
group {
    counter-reset: item;
}

item {
    counter-increment: item;
}

item::before {
    content: counter(item) " (" counters(item, ".") ") ";
}
```

The first two items display `1 (1)` and `2 (2)`. The nested items display `1 (2.1)` and `2 (2.2)`. In each function, the optional `<counter-style>` selects how each number is represented. If omitted, the style is `decimal`. For example, `counters(item, ".", upper-roman)` joins values formatted as uppercase Roman numerals. A `<counter-style>` can be the name of a predefined or author-defined `@counter-style`, or a `symbols()` function.

If the named counter does not exist at the point of use, CSS first creates it with a value of `0`. The counter functions return formatted text; they do not substitute the source element's contents.

The `content()` function takes one optional keyword:

- `text` reads the element's text, treating whitespace as if `white-space: normal` applied. This is the default when the argument is omitted.
- `before` reads the text of the element's `::before` pseudo-element.
- `after` reads the text of its `::after` pseudo-element.
- `first-letter` reads the first typographic letter, following the rules for `::first-letter`.

Read element text and attributes

CSS GCPM 3 §1.1.1.1, The content() function; CSS Values 5 §7.7, The attr() notation.

Use `content(text)` to save an element's text from the XML tree, including text inside descendant elements, in document order. Whitespace is handled as if `white-space: normal` applied. Text generated by `::before` or `::after` is not part of this value. Use `content(before)` or `content(after)` to read the corresponding pseudo-element's generated text instead.

Use `attr(name)` to save an attribute from the element matched by the rule. Without a syntax, CSS treats the attribute's literal value as a string. If the attribute is missing, the value is an empty string unless you provide a fallback. A fallback can supply text for missing or unparseable attribute values.

This sample XML has a `title` element, a `code` attribute, and an `em` child element. These names are examples; use the names and attributes present in your own XML. A CSS selector containing just `title` matches each `title` element. A selector such as `title::before` targets generated content before that element's XML text.

```
<title code="A-4">Install <em>the package</em></title>
```

```
title {
    string-set: title-text content(text), title-prefix content(before), title-
code attr(code, "No code");
}

title::before {
    content: "Chapter ";
}

@page {
    @top-left {
        content: string(title-prefix) string(title-code);
    }
    @top-right {
        content: string(title-text);
    }
}
```

Inside the `title` rule, the three comma-separated assignments save three values. `title-text` is the name chosen for the element's text. `title-prefix` saves what the `::before` rule generates. `title-code` saves the `code` attribute, or "No code" when that attribute is missing. `attr(code)` reads from the element selected by `title`, not a child or ancestor.

The @page rule applies settings to pages in the PDF. The nested `@top-left` and `@top-right` rules put content in the left and right positions of the top page margin. Their `content` declarations retrieve the saved values with `string(name)`. The left margin displays "Chapter A-4"; the right margin displays "Install the package". For a title without `code`, the left margin displays "Chapter No code".

A named string's value is assigned when the element's content box is created, or would be created if `display: none` applies. The values can change as matching elements occur in the document. The layout process must retain these successive assignments because `string()` can request a value from a particular point on a page.

13. Retrieve a named string with `string()`

    CSS GCPM 3 §1.1.2, The string() function; CSS Paged Media 3 §4.1, The @page rule; CSS Paged Media 3 §5.1, At-rules for page-margin boxes.

    Use `string(<custom-ident>, <mode>?)` in a content value to retrieve a named string. The first argument is the name you wrote in `string-set`. The optional second argument selects which assignment to use on the page. For example, `chapter-title` below must match the name after `string-set` in the earlier example, or the name you used in your own stylesheet.

    ```
    @page {
        @top-center {
            content: string(chapter-title, first);
        }
    }
    ```

    The mode accepts these four keywords:

    - `first` selects the first assignment on the page. If there is no assignment there, it uses the value in effect at the end of the previous page. This is the default.

- `start` selects the value in effect at the beginning of the page. If the named string has not yet been assigned, this value is empty.
- `last` selects the value in effect at the end of the page.
- `first-except` works like `first`, but returns an empty value on a page that contains an assignment to the named string.

The @page rule describes the page being produced. `@top-center` puts this content in the center of the top page margin.

Links

An XML document can contain text that should point to another document or to a place in the same document. HTML has standard elements and attributes for links. XML can use any element and attribute names, so CSSMedia cannot know which elements are links by looking at their names. Use the CSSMedia `-cssmedia-link` property to tell it which elements act as links and what destination each one has.

About this task

To write a link rule, first find the relevant element and attribute in your XML. The selector at the start of a rule chooses XML elements. A declaration inside the braces sets a property in the form `property: value;`. Names such as `xref` and `target` below are examples. Replace them with names used in your XML. See Selectors overview for more ways to choose elements.

Procedure

1. Set a link destination

   The `-cssmedia-link` property is a CSSMedia extension. It is not a standard CSS property, so its name has the vendor prefix `-cssmedia-`.

   CSSMedia accepts these values:

   - none means the element is not a link. This is the initial value.
   - One or more `<string>` values. A string can be a full URL, a relative address, or a fragment such as `#details`.
   - One or more `<url>` values. CSS Values 4 defines `<url>` as `url()` or `src()`.

   A `<string>` is text written between quotation marks, such as `"chapter-2.xml#installation"`. `url()` and `src()` are functions that supply a URL value.

   Strings and URLs can be mixed in one value. CSSMedia joins them in order without adding spaces or other characters. The joined result is the link destination.

   The property is not inherited. Set it on each element that should act as a link. `-cssmedia-link: none` removes the link from a selected element.

   Suppose your XML contains an `xref` element with a `target` attribute:

   ```
   <xref target="chapter-2.xml#installation">Installation instructions</xref>
   ```

   Use the element name in the selector and the attribute name in `attr()`:

   ```
   xref {
       -cssmedia-link: attr(target);
   }
   ```

   `xref` selects each element named `xref`. `attr(target)` reads the value of that element's `target` attribute. The quoted text inside the XML element is its display text; the attribute supplies its destination. If the target attribute is missing, `attr()` supplies an empty string unless you provide a fallback.

A same-document link can use a fragment identifier. This example reads the `id` attribute from an `xref` and adds `#` before it:

```
<xref target="details">More details</xref>
```

```
xref {
    -cssmedia-link: "#" attr(target);
}
```

The two parts are joined to form `#details`. The fragment identifies a location in the current XML document.

You can also write a destination directly. This example uses an attribute selector to match an `xref` whose `role` attribute is `external`:

```
<xref role="external" target="https://example.org/manual/
installation.html">Installation</xref>
```

```
xref[role="external"] {
    -cssmedia-link: "https://example.org/manual/installation.html";
}
```

A relative string destination, such as `"chapter-2.xml#installation"`, is resolved against the XML document's URL. A URL written with `url("chapter-2.xml#installation")` is resolved against the stylesheet's URL. Use a quoted string when the address is meant to be relative to the XML document.

`src()` is also part of the CSS `<url>` grammar. CSSMedia resolves its URL against the stylesheet URL, like `url()`.

5. Select links

   Selectors define these hyperlink pseudo-classes:

   - `:any-link` selects elements acting as hyperlink sources.
   - `:link` selects links. Its visited-state distinction from `:visited` applies to browsers, not PDF output.

   PDF links have no visited state. CSSMedia treats `:link` and `:any-link` identically: both select elements whose `-cssmedia-link` value is not none. The browser-history pseudo-class `:visited` is accepted by the selector parser, but never matches.

   For example, to set the color of every element that you assigned a destination:

   ```
   :any-link {
       color: navy;
   }
   ```

   The pseudo-class selects the link. The `color` declaration sets its text color. Setting the `-cssmedia-link` property alone does not choose how the link looks.

8. Select links by destination

   :local-link selects a link based on how its destination compares with the XML document's URL.

- `:local-link` matches when the destination is the same document URL. CSSMedia removes the fragment identifiers before comparing the URLs. Thus a link to `#details` is local to the current document.
- `:local-link(0)` matches links with the same URL origin. An origin consists of the URL scheme, host and port.
- `:local-link(1)` matches links with the same origin and first path segment.
- `:local-link(2)` matches links with the same origin and first two path segments. Higher non-negative integers check that many initial path segments.

For example, the following XML link points to a location in the same document:

```
<xref target="#details">More details</xref>
```

```
xref {
    -cssmedia-link: attr(target);
}

xref:local-link {
    color: navy;
}
```

The first rule gives each `xref` its destination. The second selects an `xref` whose destination is local to the current document and sets its text color. You can combine `:local-link(0)` with `:not()` to select links to other origins:

```
xref:not(:local-link(0)) {
    color: maroon;
}
```

This uses the negation pseudo-class to match `xref` links whose destination does not share the current document's origin.

Footnotes

Text and fonts

The font properties in your stylesheet decide how text looks in the PDF.

Choose a family, size and weight first. Load your own fonts when you need ones that are not available, and use font features for finer control. These topics explain each step:

- Font selection and sizing
- Font resources and descriptors
- Font variants and features
- Variable and color fonts

Font selection and sizing

Font declarations choose a family, a size and a face within that family. They can also control line spacing and whether the processor may create a style that the font does not contain. Put declarations inside a rule that selects the XML elements you want to style:

About this task

```
chapter {
    font-family: "Noto Serif", serif;
    font-size: 12pt;
    font-weight: 400;
}
```

For variant and OpenType feature properties, see Font variants and features. For variable fonts and color palettes, see Variable and color fonts.

Procedure

1. Choosing the font family and size

   `font-family`

   CSS Fonts 4 §2.1, Font family: the font-family property; CSS Fonts 4 §2.1.5, Generic font families.

   Use `font-family` to name the typeface family. Separate alternatives with commas. Put a family name containing spaces in quotes. If a family is unavailable or cannot provide a character, the processor can try the next family in the list.

   ```
   body { font-family: "Noto Serif", "Times New Roman", serif; }
   ```

   The final `serif` is a generic family name. It describes a broad typeface style rather than naming a font file. Other generic names include `sans-serif` and `monospace`. Available family names depend on the fonts CSSMedia can access.

   PDF standard fonts

   ISO 32000-1 §9.6.2, Standard Type 1 Fonts.

   PDF defines 14 standard typefaces. They are grouped here by design and face:

Group	Standard typefaces
Serif	`Times-Roman`, `Times-Bold`, `Times-Italic`, `Times-BoldItalic`
Sans-serif	`Helvetica`, `Helvetica-Bold`, `Helvetica-Oblique`, `Helvetica-BoldOblique`
Monospaced	`Courier`, `Courier-Bold`, `Courier-Oblique`, `Courier-BoldOblique`
Symbol	`Symbol`
Dingbats	`ZapfDingbats`

- The Times faces have serifs. The Helvetica faces do not. The Courier faces use fixed-width advances, so each glyph occupies the same width.
- `Symbol` contains a symbol character set. `ZapfDingbats` contains ornamental marks and symbols.
- In each four-face group, `Bold` identifies the heavier face, `Italic` the italic face, and `Oblique` the slanted face. `Times-BoldItalic` combines bold and italic; `Helvetica-BoldOblique` and `Courier-BoldOblique` combine bold and oblique.
- CSSMedia includes these standard font faces as font data. Its built-in fallback for the generic family `serif` uses the four Times faces, `sans-serif` uses the four Helvetica faces, and `monospace` uses the four Courier faces.
- CSS family lookup also includes the standard groups `Times`, `Helvetica`, `Courier`, `Symbol` and `ZapfDingbats`. The generic keywords describe broad typographic categories; the standard group names identify the bundled font families.

`font-size`

CSS Fonts 4 §2.5, Font size: the font-size property.

Use `font-size` to set the size of text. You can use a length such as `12pt`, a percentage, a relative keyword such as `larger`, or a length relative to another font size.

```
body { font-size: 12pt; }
h1 { font-size: 1.5em; }
small { font-size: smaller; }
```

An `em` value is based on the font size inherited by the selected element. A percentage is based on the parent element's font size. Use a length such as `pt` when you want to state an explicit size.

`font-size-adjust`

CSS Fonts 5 §3.4, Relative sizing: the font-size-adjust property.

Use `font-size-adjust` to request a consistent size for a chosen font measurement, such as the height of lowercase letters. This can help keep text visually similar when the preferred font is unavailable and a fallback font has different proportions.

```
body { font-size-adjust: 0.5; }
```

The property can use a metric name such as `ex-height` or `cap-height`, followed by a number, or `from-font` to use the metric from the font.

7. Selecting a font face

   `font-weight`

   CSS Fonts 4 §2.2, Font weight: the font-weight property.

   Use `font-weight` to choose how heavy the letters look. The common values are `normal` and `bold`. You can also use a number from `1` to `1000`. In the usual scale, `400` is normal and `700` is bold.

   ```
   body { font-weight: 400; }
   strong { font-weight: 700; }
   ```

   The values `bolder` and `lighter` choose a weight relative to the inherited weight. A font family may not contain every requested weight. CSSMedia uses this property when matching an available font face.

   `font-width`

   CSS Fonts 4 §2.3, Font width: the font-width property.

   Use `font-width` to request narrower or wider letters. You can use a percentage or a keyword from `ultra-condensed` through `ultra-expanded`.

   ```
   heading { font-width: condensed; }
   label { font-width: 110%; }
   ```

   The older name `font-stretch` is accepted by CSSMedia as an alias for `font-width`. A font family may not provide every width. CSSMedia uses the requested width when matching a font face.

   `font-style`

   CSS Fonts 4 §2.4, Font style: the font-style property.

   Use `font-style` to request upright or slanted letters. The usual values are `normal`, `italic` and `oblique`.

   ```
   body { font-style: normal; }
   em { font-style: italic; }
   ```

   An `oblique` value can include an angle. The specification also defines `left` and `right` oblique directions. A font family may not contain a face for every style. CSSMedia uses this property when matching a font face.

   `font-synthesis`

   CSS Fonts 4 §2.8, Controlling synthetic faces.

   If a font family does not contain a requested face, a processor may create a synthetic version, such as making regular letters look heavier. Use `font-synthesis` to allow or disable these substitutions.

   ```
   body { font-synthesis: none; }
   ```

none disables synthetic weight, style, small-caps and subscript or superscript faces. The keyword `weight`, `style`, `small-caps` or `position` allows that kind of synthesis. The longhand properties control those choices separately.

`font-synthesis-weight`

CSS Fonts 4 §2.8.1, Controlling synthesized bold: The font-synthesis-weight property.

This property controls whether the processor may make a font look heavier when the family lacks the requested weight. Use none to forbid that synthetic bolding, or `auto` to allow it.

`font-synthesis-style`

CSS Fonts 4 §2.8.2, Controlling synthesized oblique: The font-synthesis-style property.

This property controls whether the processor may slant a font when the family lacks the requested italic or oblique face. Use none to forbid a synthetic slant. `oblique-only` allows synthesis for oblique styles but not italic styles. `auto` allows the normal synthesis behavior.

`font-synthesis-small-caps`

CSS Fonts 4 §2.8.3, Controlling synthesized small caps: The font-synthesis-small-caps property.

This property controls whether the processor may create small capitals when the font does not provide them. Use none to forbid synthetic small capitals, or `auto` to allow them.

`font-synthesis-position`

CSS Fonts 4 §2.8.4, Controlling synthesized super- and subscripts: The font-synthesis-position property.

This property controls whether the processor may create subscript or superscript forms when the font does not provide them. Use none to forbid those synthetic forms, or `auto` to allow them.

8. Shorthand and line spacing

   `font`

   CSS Fonts 4 §2.7, Shorthand font property: the font property.

   Use `font` to set several font properties in one declaration. Write optional style, variant, weight and width values first. Then write the required size, an optional slash and line height, and the required family.

   ```
   chapter { font: italic bold 12pt/1.4 "Noto Serif", serif; }
   ```

   The size and family are required. The shorthand also accepts system font keywords such as `caption` and `menu`. It resets other properties in the shorthand to their initial values. Use separate declarations when you want to set advanced font features without resetting them.

   CSS Fonts 4 allows a user agent to substitute its default font when the requested system font is unavailable. CSSMedia uses the `system-ui` family and initial values for the other font subproperties as that fallback. It does not resolve platform-specific settings for roles such as `caption` or `menu`.

   `line-height`

   CSS Inline Layout 3 §5.1, Line Spacing: the line-height property.

Use `line-height` to control the height of a line of text. A unitless number multiplies the font size. A length states the line height directly. A percentage is relative to the font size. `normal` lets the processor choose a value.

```
body { font-size: 12pt; line-height: 1.4; }
caption { line-height: 16pt; }
```

With a font size of `12pt`, `line-height: 1.4` requests a line height of 1.4 times the font size. A unitless number scales when the font size changes.

Font resources and descriptors

Use `@font-face` to make a font face available to a stylesheet. The rule names the family, identifies one or more sources and describes the face so the font matcher can choose it.

About this task

CSS Fonts 4 §4.1, The `@font-face` rule.

```
@font-face {
    font-family: "Example Serif";
    src: url("example-serif.woff2") format("woff2");
    font-weight: 400 700;
    font-style: normal;
    unicode-range: U+0000-00FF;
}

chapter {
    font-family: "Example Serif", serif;
}
```

The descriptors inside the rule describe the font face, not the selected XML element. Use separate rules for separate faces in the family, such as regular, bold and italic faces. A rule needs valid `font-family` and `src` descriptors. Invalid declarations are ignored. When a descriptor occurs more than once, the last valid value is used.

Procedure

1. Identify the family and source

   `font-family`

   CSS Fonts 4 §4.2, Font family descriptor.

   This descriptor gives the face the family name used by a `font-family` property. It replaces family names stored in the font file. Write the name as a quoted string or as one or more unquoted identifiers. Separate words in an unquoted name with spaces. A name that is a CSS-wide keyword must be quoted. A declared family name also takes precedence over an installed family with the same name.

   `src`

   CSS Fonts 4 §4.3, Font reference descriptor; parsing; source selection; CSS Fonts 5 `src` descriptor.

This descriptor is a comma-separated, ordered list of sources. Each source is either a URL with optional hints or a `local()` reference. When choosing a source, a processor checks the entries in order and can skip entries whose declared format or technology it does not support.

```
@font-face {
    font-family: "Example Serif";
    src: local("Example Serif Regular"),
         url("example-serif.woff2") format("woff2"),
         url("example-serif.otf") format("opentype");
}
```

`url()` identifies a font resource. A relative URL is resolved against the stylesheet's base URL. `local()` identifies an installed font by its full name or PostScript name. It does not refer to a file. The processor tries sources in order and proceeds to the next source if it cannot find or use the current one.

`format()`

CSS Fonts 4 `format()` hint; font format definitions.

The optional `format()` hint identifies a URL resource's format. Use one format keyword or a quoted format string.

- `collection` identifies a TrueType or OpenType font collection, which can contain multiple font faces.
- `embedded-opentype` identifies Embedded OpenType resources.
- `opentype` identifies OpenType resources. It does not promise a particular glyph outline format or that the font contains OpenType layout tables.
- `svg` identifies an SVG font.
- `truetype` identifies TrueType resources.
- `woff` identifies WOFF 1.0 resources.
- `woff2` identifies WOFF 2.0 resources.

The `opentype` and `truetype` hints are treated as synonyms by CSS Fonts. The hint lets a processor avoid a resource format it cannot use. It does not convert the resource.

For compatibility with older stylesheets, CSS Fonts 4 also accepts these quoted forms: `format("woff2")`, `format("woff")`, `format("truetype")`, `format("opentype")` and `format("collection")`. They mean the corresponding unquoted format keywords.

The legacy strings `"woff2-variations"`, `"woff-variations"`, `"truetype-variations"` and `"opentype-variations"` mean the corresponding format keyword followed by `tech(variations)`.

`tech()`

CSS Fonts 4 `tech()` hint; font technology definitions.

The optional `tech()` function lists technologies required by the resource. Separate multiple values with commas. The processor can skip the source if it does not support any listed requirement.

- `features-opentype` , `features-aat` and `features-graphite` indicate support for OpenType, Apple Advanced Typography or Graphite layout features, respectively.
- `color-COLRv0` and `color-COLRv1` indicate color glyphs in the corresponding COLR table versions.
- `color-SVG` indicates SVG color glyphs.
- `color-sbix` indicates the sbix bitmap color glyph format.
- `color-CBDT` indicates the CBDT bitmap color glyph format.
- `variations` indicates support for font variation axes.
- `palettes` indicates color-font palette support.
- `incremental` indicates support for incremental font transfer.
- `avar2` indicates support for OpenType version 2 axis variations, which remap variable-font axes.

`format()` precedes `tech()` when a source uses both. The four `-variations` strings listed above each mean the corresponding format hint plus `tech(variations)`.

18. Describe which face the source provides

    `font-weight`

    CSS Fonts 4 font property descriptors; weight values.

    This descriptor tells the font matcher which weight or weight range the face provides. `auto` uses the weight information in the font. Otherwise, provide one absolute weight or two ascending endpoints. Two endpoints include every weight in the interval.

    - `normal` is `400`.
    - `bold` is `700`.
    - A number from `1` through `1000` gives the weight directly. Larger numbers mean heavier weights.

    The descriptor's range tells the matcher which requested weights this face can satisfy. It does not change the font's data.

    `font-width`

    CSS Fonts 4 font property descriptors; width values.

    This descriptor tells the matcher which width or width range the face provides. `auto` uses the width information in the font. Otherwise, provide one value or two ascending endpoints. A non-negative percentage represents glyph width relative to normal width. Values below `100%` are narrower; values above `100%` are wider.

    The named values correspond to these percentages:

    - `ultra-condensed` : `50%`.
    - `extra-condensed` : `62.5%`.
    - `condensed` : `75%`.
    - `semi-condensed` : `87.5%`.
    - `normal` : `100%`.
    - `semi-expanded` : `112.5%`.

- `expanded` : `125%`.
- `extra-expanded` : `150%`.
- `ultra-expanded` : `200%`.

The legacy descriptor name `font-stretch` is an alias for `font-width`.

`font-style`

CSS Fonts 4 font property descriptors; style values.

This descriptor tells the matcher which style or style range the face provides. `auto` uses the style information in the font.

- `normal` describes an upright face with no slant.
- `italic` describes an italic face. If one is unavailable, matching can use an oblique face.
- `left` describes an italic or oblique face with a positive, clockwise slant.
- `right` describes an italic or oblique face with a negative, counter-clockwise slant.
- `oblique` describes a slanted face. It can take one angle or two ascending endpoints from `-90deg` to `90deg`. With no angle, the value is `14deg`. Positive angles are clockwise; negative angles are counter-clockwise.

Two values define an inclusive angle range. Fractional angles are allowed.

`font-size`

CSS Fonts 5 font size descriptor.

This descriptor limits the font sizes for which the face is a match. `auto` matches any size. One number matches that size; two numbers define an inclusive range. The values are unitless numbers, not CSS lengths.

`font-named-instance`

CSS Fonts 4 `font-named-instance` descriptor.

This descriptor selects a named instance in a variable font. `auto` leaves instance selection to the normal font variation process. A string selects the first named instance whose localized name matches the string. If no such instance exists, the descriptor behaves as `auto`. The requested `font-weight`, `font-width`, and `font-style` variation values are applied before the named instance's axis values.

36. Limit a face to characters

    `unicode-range`

    CSS Fonts 4 character range descriptor.

    This descriptor gives the Unicode code points for which the face is eligible. Its value is a comma-separated list:

    - A single code point has a form such as `U+0041`.
    - A range has a form such as `U+0400-04FF`. Both endpoints are included.
    - A wildcard uses one or more trailing question marks, as in `U+4??`. Each question mark stands for one hexadecimal digit.

    Hexadecimal digits are ASCII case-insensitive. Endpoints must be valid Unicode code points, and a range's end cannot precede its start. The descriptor filters the characters for which

the face can be considered. It does not guarantee that the font contains a glyph for each one. The default range includes all Unicode code points.

40. Set face-specific font behavior

    `font-display`

    CSS Fonts 4 font display descriptor.

    This descriptor controls the font's loading behavior while text is being displayed. The processor divides loading into a block period, when text may be invisible, and a swap period, when a loaded font may replace the fallback.

    - `auto` lets the processor choose the policy.
    - `block` uses an invisible-text block period, then allows the downloaded font to replace the fallback.
    - `swap` shows fallback text immediately and allows the downloaded font to replace it.
    - `fallback` allows a short block period and a limited swap period. A late font load does not replace the fallback.
    - `optional` allows a very short block period and no swap period. The processor may use the font only when it is available promptly.

    The user agent determines the durations of the block and swap periods.

    `font-language-override`

    CSS Fonts 4 font language override descriptor.

    This descriptor sets the OpenType language-system tag used for the face's language-sensitive features. `normal` uses the language selected for the text. A string supplies a case-sensitive OpenType language-system tag, such as `"SRB"`. Find tags in the font's documentation or in Microsoft's OpenType language system tags. These are not BCP 47 tags such as `sr` or `sr-Latn`; use BCP 47 tags in `xml:lang` to identify the text's language. The OpenType tag is matched as four characters; shorter strings are padded at the end with U+0020 spaces. Unknown tags are ignored.

    `font-feature-settings`

    CSS Fonts 4 font feature settings descriptor; feature tag values.

    This descriptor sets initial low-level OpenType feature settings for the face. It does not affect font selection. `normal` makes no explicit change to glyph selection or positioning. Otherwise, give a comma-separated list of feature tags. Each tag is a case-sensitive string of exactly four printable ASCII characters. A tag can be followed by an integer, `on` or `off` . If the value is omitted, it is `1`. on means `1`; `off` means `0`. `0` disables the feature. For a boolean feature, `1` enables it. For a non-boolean feature, an integer of `1` or greater enables it and selects that feature index. Tags that the font does not contain are ignored. See the feature tag value syntax.

    `font-variation-settings`

    CSS Fonts 4 font variation settings descriptor.

    This descriptor sets initial low-level variation-axis settings for the face. It does not affect font selection. `normal` does not request additional axis settings. Otherwise, give a comma-separated list of case-sensitive axis tags paired with numbers, such as `"wght" 650`. An axis

tag is a string of exactly four printable ASCII characters. Each number is an axis value; values outside the range supported by the font are clamped to that range. An axis tag the font does not contain is ignored. See the `font-variation-settings` property for the axis-tag rules. Prefer `font-weight`, `font-width` or `font-style` for the standard weight, width and slant axes. Those properties express the requested font characteristics and can also match non-variable faces.

46. Override font metrics

    `size-adjust`

    CSS Fonts 5 glyph size multiplier descriptor.

    This descriptor scales a face's glyph outlines and associated metrics by a non-negative percentage. `100%` leaves them unchanged. Other values scale glyphs, advances and metrics together. The CSS `font-size` remains unchanged.

    `ascent-override` , `descent-override` and `line-gap-override`

    CSS Fonts 5 font metrics override descriptors.

    `ascent-override` replaces the ascent metric. `descent-override` replaces the descent metric. `line-gap-override` replaces the line-gap metric. Each accepts `normal` or a non-negative percentage. The percentage replaces the corresponding metric with that percentage of the effective font size after adjustment. `normal` uses the metric from the font as if the descriptor were absent. Each descriptor accepts one value for the x-axis and an optional second value for the y-axis. The second value defaults to `normal`.

    Superscript and subscript metric overrides

    CSS Fonts 5 superscript and subscript metric override descriptors.

    `superscript-position-override` and `subscript-position-override` replace the corresponding glyph position metrics. `superscript-size-override` and `subscript-size-override` replace their glyph size metrics. These metrics are used when `font-variant-position` synthesizes superscript or subscript glyphs.

    Each descriptor accepts `normal`, `from-font` or a percentage. For a position descriptor, the percentage can be positive or negative. For a size descriptor, it cannot be negative.

    - `normal` lets the processor choose the metric from the font or a heuristic.
    - `from-font` uses the corresponding font metric when present. If it is absent, it behaves as `normal`.
    - A percentage replaces the metric with that percentage of the used font size.

    The descriptors accept one value for the x-axis and an optional second value for the y-axis. If the second value is omitted, the x-axis value is used. The y-axis metric is used for upright text in vertical typographic modes.

Font variants and features

Fonts can contain alternate letter shapes, number styles and ligatures. Use the properties in this guide to request those forms. The result depends on the selected font and its `font-synthesis` settings. See Font selection and sizing for the synthesis properties and their values.

Procedure

1. Common font features

   `font-kerning`

   CSS Fonts 4 §6.3, Kerning: the font-kerning property.

   Kerning adjusts the space between particular pairs of letters when the font provides kerning data. Use `normal` to request kerning, `none` to turn it off, or `auto` to let the processor decide.

   ```
   body { font-kerning: normal; }
   ```

   `font-variant-ligatures`

   CSS Fonts 4 §6.4, Ligatures: the font-variant-ligatures property.

   Ligatures replace certain letter combinations with a single designed glyph. Use this property to control common, discretionary, historical or contextual ligatures.

   ```
   body { font-variant-ligatures: common-ligatures; }
   code { font-variant-ligatures: none; }
   ```

   `normal` uses the default feature behavior. For OpenType fonts, common ligatures and contextual forms are on by default. Discretionary and historical ligatures are off by default. `none` disables all kinds of ligatures and contextual forms covered by this property.

   - `common-ligatures` enables common ligatures, such as `fi` or `fl`, when the font provides them. `no-common-ligatures` disables them.
   - `discretionary-ligatures` enables optional ligatures intended for particular styles. `no-discretionary-ligatures` disables them.
   - `historical-ligatures` enables historical ligatures. `no-historical-ligatures` disables them.
   - `contextual` enables contextual forms, where a glyph changes according to nearby characters. `no-contextual` disables them.

   You can combine one value from each pair. Required ligatures used to render complex scripts are not affected by these settings.

   `font-variant-caps`

   CSS Fonts 4 §6.6, Capitalization: the font-variant-caps property.

   Use `font-variant-caps` to request small capitals or other capitalization forms from a font.

   ```
   term { font-variant-caps: small-caps; }
   ```

   `normal` uses the usual letter forms. `small-caps` uses small capitals for lowercase letters. `all-small-caps` uses small capitals for both lowercase and uppercase letters. `petite-caps` uses petite capitals for lowercase letters. `all-petite-caps` uses petite capitals for both cases. `unicase` uses small capitals for uppercase letters with ordinary lowercase letters. `titling-caps` requests capitals designed for titles.

If the font does not provide the requested forms, `font-synthesis-small-caps` determines whether the processor synthesizes them.

`font-variant-position`

CSS Fonts 4 §6.5, Subscript and superscript forms: the font-variant-position property.

Use `font-variant-position` to request subscript or superscript glyph forms from the font.

```
formula { font-variant-position: sub; }
```

`normal` uses the ordinary glyphs. `sub` requests subscript glyphs. `super` requests superscript glyphs. The alternate glyphs use the same em-box and baseline as ordinary glyphs, so this property does not resize or reposition them. It does not change the meaning or position of the XML element. If the font does not provide the requested forms, `font-synthesis-position` determines whether the processor synthesizes them.

`font-variant-numeric`

CSS Fonts 4 §6.7, Numerical formatting: the font-variant-numeric property.

Use this property to request number shapes, spacing and fractions. `normal` uses the font's default forms. You may combine one value from each mutually exclusive pair:

```
table { font-variant-numeric: tabular-nums; }
```

- `lining-nums` requests figures aligned with the capital-letter height. `oldstyle-nums` requests figures with ascenders and descenders.
- `proportional-nums` requests figures whose widths vary. `tabular-nums` requests figures with equal widths.
- `diagonal-fractions` requests a numerator and denominator separated by a diagonal slash. `stacked-fractions` requests a numerator above a denominator.
- `ordinal` requests letter forms used with ordinal numbers, such as the suffix in `1st`. `slashed-zero` requests a zero glyph with a slash.

The font must provide each requested feature.

10. Alternate glyphs and language forms

    `font-variant-alternates`

    CSS Fonts 4 §6.8, Alternates and swashes: the font-variant-alternates property.

    Use `normal` to request no named alternate. Otherwise, combine one or more alternate requests:

    ```
    heading { font-variant-alternates: historical-forms; }
    ```

    `stylistic(name)` requests one named stylistic alternate. `historical-forms` requests historical glyph forms. `styleset(name, ...)` requests one or more named stylistic sets. `character-variant(name, ...)` requests named variants for individual characters. `swash(name)` requests a named swash form. `ornaments(name)` requests a named ornament. `annotation(name)` requests a named annotation form.

Names in these functions are case-sensitive. A name can be defined for a font family by `@font-feature-values`. If it is not defined for the selected family and feature type, it has no effect. The font must support the mapped feature.

`@font-feature-values`

CSS Fonts 4 §6.9, Defining font specific alternates: the @font-feature-values rule.

Use `@font-feature-values` to give readable names to alternate features that a font identifies by OpenType feature tags. Its prelude is one or more named font families separated by commas. Generic and system font names are not allowed. Inside the block, use feature-value rules to map names to the font-specific integer indexes.

```
@font-feature-values "Example Serif" {
    @styleset {
        headings: 1;
    }
}

h1 { font-variant-alternates: styleset(headings); }
```

Feature-value rules

CSS Fonts 4 §6.9.1, Basic syntax.

Each nested rule has a declaration block. A declaration maps a case-sensitive name to a non-negative integer or integer list. The meaning and number of integers depend on the rule:

- `@stylistic` maps a name to one stylistic-alternate index for OpenType feature `salt`.
- `@historical-forms` maps a name to one or more historical-form indexes for OpenType feature `hist`.
- `@styleset` maps a name to one or more style-set indexes. Values from `1` through `99` select the corresponding `ss01` through `ss99` features. The OpenType standard defines `ss01` through `ss20`; later feature tags may be used by fonts that support them. The value `0` and values above `99` are syntactically valid but enable no OpenType feature. See CSS Fonts 4 §6.9.2.
- `@character-variant` maps a name to one or two indexes. The first selects `cv01` through `cv99`; the optional second sets that feature's variant value. A first index of `0` or greater than `99` is valid but enables no feature. See CSS Fonts 4 §6.9.2.
- `@swash` maps a name to one swash index for OpenType feature `swsh` or `cswh`.
- `@ornaments` maps a name to one ornament index for OpenType feature `ornm`.
- `@annotation` maps a name to one annotation index for OpenType feature `nalt`.

Repeated feature-value rules of the same type contribute their declarations together. If the same family, rule type and name are defined more than once, the last definition is used.

`@font-feature-values` also accepts a `font-display` descriptor. Its initial value is `auto` , which leaves the policy to the processor. The descriptor sets the default font-display policy for `@font-face` rules for the listed family when those rules do not set their own `font-display`. `block` uses a short block period, with 3 seconds recommended, followed by an infinite swap period. `swap` uses an extremely short block period, with 100 milliseconds or less recommended, followed by an infinite swap period. `fallback` uses an extremely short

block period, with 100 milliseconds or less recommended, followed by a short swap period, with 3 seconds recommended. `optional` uses the font only if it is available for the first rendering. Otherwise, the block and swap periods are treated as expired, so the fallback remains in use.

`font-variant-east-asian`

CSS Fonts 4 §6.10, East Asian text rendering: the font-variant-east-asian property.

Use `normal` to request none of the variants below. You can combine a glyph-variant value, a width value and `ruby`:

```
ruby { font-variant-east-asian: ruby; }
```

- `jis78` , `jis83` , `jis90` and `jis04` request Japanese glyph forms from the corresponding JIS standard editions.
- `simplified` requests simplified-character forms. `traditional` requests traditional-character forms.
- `full-width` requests full-width glyphs. `proportional-width` requests proportionally spaced glyphs.
- `ruby` requests glyph forms designed for ruby text.

The font must provide the requested forms. Ruby-specific glyphs are designed to remain legible at the smaller sizes used for ruby text.

`font-language-override`

CSS Fonts 4 §6.13, Font language override: the font-language-override property.

Set the text's language with a BCP 47 tag in `xml:lang`. For example, `tr` identifies Turkish and `en-GB` identifies British English. Use a language tag that follows BCP 47 and its registered subtags in the IANA Language Subtag Registry.

Fonts can use OpenType language-system tags to select language-specific glyph features. Normally, the text's BCP 47 language lets the processor choose the corresponding OpenType system. Set `font-language-override` only when you need to select a different OpenType system, such as when the font lacks the system for the text's language.

```
chapter { font-language-override: "SRB"; }
```

For example, use `xml:lang="mk"` to identify Macedonian text. If the selected font lacks a Macedonian language system but provides Serbian conventions that suit the text, `"SRB"` selects its Serbian OpenType system.

`normal` uses the OpenType system inferred from the text's BCP 47 language. A `string` must be a case-sensitive OpenType language-system tag, not a BCP 47 tag. Find the tag in the font's documentation or in Microsoft's OpenType language system tags. The tag is matched as four characters; shorter strings are padded at the end with U+0020 spaces. Unknown tags are ignored. Leave the value as `normal` unless you need to override the inferred system.

`font-variant-emoji`

CSS Fonts 4 §9.3, Selecting the text presentation style: The font-variant-emoji property.

`normal` lets the processor choose text or emoji presentation according to its usual policy. `text` treats each character that can have emoji presentation as if followed by U+FE0E, the text presentation selector. `emoji` treats each such character as if followed by U+FE0F, the emoji presentation selector. `unicode` follows the Unicode default presentation for each character. An explicit text or emoji variation selector in the text overrides that default.

```
label { font-variant-emoji: text; }
```

The selected font must contain the requested glyphs and presentation.

22. Shorthand and low-level control

    `font-variant`

    CSS Fonts 4 §6.11, Overall shorthand for font rendering: the font-variant property.

    Use `font-variant` to set several variant properties in one declaration. `normal` resets every subproperty to its initial value. `none` sets `font-variant-ligatures` to none and resets the other variant subproperties. Otherwise, combine compatible values from the ligature, capitalization, alternate, numeric, East Asian, position and emoji groups described above. Omitted subproperties reset to their initial values.

    ```
    body { font-variant: small-caps; }
    table { font-variant: lining-nums tabular-nums; }
    ```

    Use a longhand such as `font-variant-numeric` when you want to control only one kind of feature. The `font` shorthand is different. It resets font variant settings rather than setting all these detailed choices. `font-variant` does not reset `font-language-override`, `font-feature-settings` or `font-variation-settings`.

    `font-feature-settings`

    CSS Fonts 4 §6.12, Low-level font feature settings control: the font-feature-settings property.

    Use `font-feature-settings` to set a specific OpenType feature directly. Write the feature tag as a four-character string, followed by an optional number, on or `off`.

    ```
    body { font-feature-settings: "liga" 0; }
    ```

    `normal` makes no change to glyph selection or positioning. Otherwise, give one or more comma-separated `<feature-tag-value>` items. Each tag is a case-sensitive string of four printable ASCII characters, from U+0020 through U+007E. The font must define the tag. Unsupported tags are ignored. This property does not synthesize their behavior, except that a processor may synthesize kerning for fonts with a legacy `kern` table.

    For each tag, omit the value or use on to enable the feature with value `1`. Use `off` or `0` to disable it. The optional integer value cannot be negative. A positive integer enables the feature and selects its indexed form when it has multiple forms. on is equivalent to `1`; `off` is equivalent to `0`. If a tag is repeated, its last value takes precedence.

    In the example, `liga` is the standard ligature feature and `0` disables it. Prefer a higher-level property such as `font-variant-ligatures` when it describes the feature you need.

Variable and color fonts

Variable fonts can change their design along axes such as weight, width or optical size. Color fonts can draw glyphs with indexed colors from a palette. The font file must provide the axis or palette that you request.

Procedure

1. Variable font controls

   `font-optical-sizing`

   CSS Fonts 4 §8.1, Optical sizing control.

   An optical-size axis lets a font adjust glyph details for different text sizes. The property accepts `auto` or `none` .

   ```
   body { font-optical-sizing: auto; }
   ```

   With `auto`, the processor may adjust glyphs for the used font size. Variable OpenType and TrueType fonts commonly do this through the `opsz` axis. The processor must stay within the axis range in the font and must not synthesize optical sizing when the font does not provide it. With none, it must not change glyph shapes for optical size.

   `font-variation-settings`

   CSS Fonts 4 §8.2, Low-level font variation settings control.

   Use this property to set one or more variable-font axes directly. Each item is a four-character, case-sensitive `<opentype-tag>` followed by a number. The tag must contain printable ASCII characters, from U+0020 through U+007E. The number may be fractional or negative.

   ```
   heading { font-variation-settings: "wght" 650, "wdth" 90; }
   ```

   `normal` means this property makes no change to glyph shape, matching or positioning. An unknown axis tag or an axis not supported by the selected font has no effect. A number outside the supported range for an axis is clamped to that range. If a tag occurs more than once, its last value is used.

   Use higher-level properties for common axes. `font-weight` controls the `wght` axis, `font-width` controls `wdth`, `font-style` controls `slnt` or `ital`, and `font-optical-sizing` controls `opsz`. These properties participate in font matching; `font-variation-settings` does not.

2. Color font palettes

   `font-palette`

   CSS Fonts 4 §9.1, Controlling color font palettes.

   Use this property to choose the colors used by a color font. It accepts `normal` , `light` , `dark` , a named palette, or `palette-mix()` .

   ```
   emoji { font-palette: dark; }
   ```

- `normal` uses the font's normal palette selection. For OpenType CPAL fonts, this is usually palette index `0`. When `color-scheme` requests a light or dark scheme, the specification says the processor should select the first matching palette.
- `light` selects the first palette marked for light backgrounds. If the font format has no such metadata, or no palette is marked, this behaves as `normal`.
- `dark` selects the first palette marked for dark backgrounds. If the font format has no such metadata, or no palette is marked, this behaves as `normal`.
- A palette identifier names an author-defined palette from `@font-palette-values`. The identifier is a `<dashed-ident>`, so it starts with two hyphens, as in `--night`. If no applicable rule defines it, it behaves as `normal`.
- `palette-mix()` makes a palette by interpolating corresponding colors from one or more other palettes. Its arguments are `font-palette` values, which may be keywords, named palettes or nested `palette-mix()` functions. Each may have a percentage from `0%` to `100%`. Omitted percentages share the remaining percentage equally. If specified percentages total more than `100%`, they are normalized; if they total less, the result's opacity is reduced by the unassigned share. The rules follow CSS Color 5 §3.2 and `color-mix()`.

For example, `palette-mix(in oklab, dark 25%, --night)` blends one quarter of the dark palette with three quarters of `--night`. The optional `<color-interpolation-method>` selects the space used to blend corresponding colors. If omitted, the default is `oklab`.

- Rectangular RGB spaces include `srgb`, `srgb-linear`, `display-p3`, `display-p3-linear`, `a98-rgb`, `prophoto-rgb` and `rec2020`. The `-linear` forms interpolate light rather than gamma-encoded channel values.
- `lab` and `oklab` use lightness and two opponent-color coordinates. The `xyz` spaces use CIE XYZ tristimulus coordinates: `xyz`, `xyz-d50` and `xyz-d65` identify the reference white.
- Polar spaces represent colors with an angle and other coordinates. `hsl` uses hue, saturation and lightness. `hwb` uses hue, whiteness and blackness. `lch` and `oklch` use lightness, chroma and hue.
- A custom color space is a dashed identifier declared by `@color-profile`.
- A polar space may also specify a hue direction using `shorter`, `longer`, `increasing` or `decreasing`. `shorter` follows the shorter route around the hue circle, `longer` the longer route, `increasing` moves toward larger hue angles, and `decreasing` moves toward smaller hue angles. If no direction is written, `shorter` is used. See the color interpolation grammar for the complete syntax.

`@font-palette-values`

CSS Fonts 4 §9.2, User-defined font color palettes.

Use this rule to give a palette a name, choose its starting palette and replace selected colors. Its name must be a `<dashed-ident>`, such as `--night`. When rules use the same name, the last rule in document order takes effect.

```
@font-palette-values --night {
    font-family: "Example Emoji";
    base-palette: dark;
    override-colors: 0 navy, 1 white;
}

emoji {
    font-family: "Example Emoji";
    font-palette: --night;
}
```

The rule requires a `font-family` descriptor. It accepts one or more named font families, separated by commas. A family name may be a string or a sequence of identifiers, such as `"Example Emoji"` or `Example Sans`. Generic family names such as `serif` are not allowed. The palette applies only to font faces whose family name matches.

`base-palette`

CSS Fonts 4 §9.2.2, Specifying the base palette.

This descriptor chooses the palette in the font on which the named palette is based.

- `light` selects the first palette marked for light backgrounds. If the font format has no such metadata, or no palette is marked, it means palette index `0`.
- `dark` selects the first palette marked for dark backgrounds. If the font format has no such metadata, or no palette is marked, it means palette index `0`.
- A non-negative integer selects a zero-based palette index. For example, `0` selects the first palette and `2` selects the third.

If this descriptor is omitted, or the font has no palette at that index, the base is palette `0`. Colors not replaced by `override-colors` remain from the base palette.

`override-colors`

CSS Fonts 4 §9.2.3, Overriding colors from a palette.

This descriptor takes a comma-separated list of palette entries and replacement colors. Each entry is a non-negative integer followed by an absolute CSS color. Palette indexes start at `0`.

```
@font-palette-values --night {
    font-family: "Example Emoji";
    base-palette: 0;
    override-colors: 0 navy, 1 rgb(255 255 255);
}
```

Each pair replaces the color at that index in the base palette. An index outside the palette's range is ignored. If an index is specified more than once, the last color is used. Every color in the descriptor must be absolute, meaning its computed value has a colorimetric meaning independent of the current text color or color mode. Named colors, hexadecimal colors, and numeric `rgb()` colors are examples. `currentColor`, system colors, `light-dark()`, `contrast-color()` and `device-cmyk()` are not absolute. A non-absolute color makes the descriptor invalid. Wrapping a relative color in `color-mix()` or relative color syntax does not make it absolute. An absolute color may use any supported color space.

Math functions, `var()` and `env()` may be used in descriptor values. They are evaluated in the root element's context. When a descriptor occurs more than once in one rule, the last valid declaration is used.

Languages and hyphenation

Document language

The language of the text decides how CSSMedia hyphenates words, which quotation marks it uses, how it changes letter case and which elements the `:lang()` selector matches. CSSMedia reads the language from the `xml:lang` attribute. If your XML names the language in another attribute, use the CSSMedia `-cssmedia-lang` property to say which one.

About this task

A language tag is a short code that names a language, such as `en` for English, `fr` for French or `zh-Hans` for Chinese in simplified characters. The `xml:lang` attribute is defined by the XML specification. It holds a language tag and applies to the element that carries it and to everything inside that element.

XML can use any element and attribute names, so CSSMedia cannot know which other attribute holds a language. A vocabulary such as DocBook 4 uses a plain `lang` attribute. Names such as `para` and `lang` below are examples. Replace them with names used in your XML.

Procedure

1. Mark the language with `xml:lang`

   When your XML already has `xml:lang` attributes, you do not write anything in the stylesheet. An element has the language of the nearest `xml:lang` on itself or on an ancestor.

   ```
   <article xml:lang="en">
       <para>This paragraph is in English.</para>
       <para xml:lang="fr">Ce paragraphe est en français.</para>
   </article>
   ```

2. Take the language from another attribute

   The `-cssmedia-lang` property is a CSSMedia extension. It is not a standard CSS property, so its name has the vendor prefix `-cssmedia-`. It sets the language of the elements it is applied to.

Value	Meaning
none	The property does not set the language of the element. This is the initial value.
`<string>`	A language tag written between quotation marks, such as `"fr"`. An empty string is the same as none.

   The property is not inherited, but the language it sets reaches the content of the element, as `xml:lang` does. Use `attr()` to read the language from an attribute:

```
<book lang="en">
    <para>This paragraph is in English.</para>
    <para lang="de">Dieser Absatz ist auf Deutsch.</para>
</book>
```

```
* {
    -cssmedia-lang: attr(lang);
}
```

`*` selects every element. `attr(lang)` reads the value of the `lang` attribute of each element. The first `para` has no `lang` attribute, so `attr()` supplies an empty string and the paragraph keeps the language of `book`. The second `para` is in German.

You can also write the language in the stylesheet:

```
quote[role="greek"] {
    -cssmedia-lang: "el";
}
```

3. Know which language wins

   For each element, CSSMedia takes the first of these that exists:

   4. The language set by `-cssmedia-lang` on the element.
   5. The `xml:lang` attribute of the element itself.
   6. The language of its parent element.

   An element with both a `lang` attribute that your rule reads and an `xml:lang` attribute takes the language from `lang`.

   A rule that sets `-cssmedia-lang` cannot depend on the language. A selector such as `:lang(fr)` or `:dir(rtl)` never matches while CSSMedia decides the language, so those selectors have no effect on the `-cssmedia-lang` declarations. They work as usual on all other properties, and they see the language that `-cssmedia-lang` gave.

Ruby and vertical text

Boxes, backgrounds and borders

Every element in your document generates a box that you can size, space and fill with color.

A box has content, padding, a border and a margin. These topics explain how to size and space boxes and how to set their backgrounds:

- Box sizing, margins and padding
- Box dimensions
- Background color and images
- Background positioning and size
- Background repetition and attachment
- Background clipping, shorthand and blending

Box sizing, margins and padding

What is a CSS box?

A CSS box is the area layout creates to place an element and its content on a page. It is not an XML tag and does not have to be drawn with a visible border. CSS uses the element's `display` value to decide whether it generates a box. Usually an element generates one box, but some display values generate none or more than one. See CSS Display 3, Box generation.

The XML element is the source that CSS selects. The box is the area that layout sizes and positions. The box's content area holds text, child boxes, images or other content. Around it, CSS can add padding, a border and margin:

- **Content** is the text or other content inside the box.
- **Padding** is space between the content and the border.
- **Border** is the edge around the padding. It is visible when border styling gives it a visible style.
- **Margin** is space outside the border, between this box and surrounding boxes.

These areas may have zero size. Their relationship is defined by CSS Box Model 4 §2, The CSS Box Model. Sizing properties set the content area's dimensions, with `box-sizing` choosing whether the declared size measures the content box or border box. Margin and padding properties set the corresponding outer and inner spaces.

The properties described here also accept the CSS-wide keywords. See Inheritance and default values.

The **containing block** is the reference box used to calculate some sizes and percentages. It is often formed by an ancestor box. Its inline size is the size along the text line. In horizontal writing, that is usually the width. See CSS Display 3, Containing Blocks and CSS Writing Modes 4, Abstract Box Terminology.

Units in size and spacing values

CSS Values 4 §5.5, Percentages; CSS Values 4 §6, Distance Units.

A length is a number followed by a unit, such as `12pt` or `1em`. The unit tells CSS how to interpret the number. A percentage, such as `60%`, is a different kind of value: it means a fraction of a

reference size. Which size it refers to depends on the property. For the sizing properties below, it is generally a corresponding size of the containing block; for margins and padding, it is based on the containing block's inline size.

Common length units used in this guide are:

Unit	Meaning
`px`	CSS pixel. CSS defines `96px` as `1in`.
`in`	Inch. `1in` equals `2.54cm` or `96px`.
`cm`	Centimeter.
`mm`	Millimeter.
`Q`	Quarter-millimeter. `4Q` equals `1mm`.
`pt`	Point. `72pt` equals `1in`.
`pc`	Pica. `6pc` equals `1in`.
`em`	Font-relative length based on the selected element's font size.
`rem`	Font-relative length based on the root element's font size.

- For example, `padding: 12pt` sets twelve points of padding on each side. `margin-block: 1em` sets a margin based on the element's font size. The same `1em` can therefore represent different lengths on elements with different font sizes.
- `60%` is not a length unit; it scales with the property's percentage reference size.
- CSS also defines other font-relative, viewport-relative and container-relative length units. See CSS Values 4 §6, Distance Units for the full list and definitions.
- A zero length may be written as `0` without a unit; nonzero lengths need a unit. The allowed range depends on the property. For example, margins may be negative, but padding cannot.

Choose what a size measures

CSS Sizing 3 §3, Specifying Box Sizes.

`box-sizing`

`box-sizing` determines whether a fixed size measures the content box or the border box.

- `content-box` measures the content area. Padding and border are added outside the specified size.
- `border-box` measures through the outside border edge. The content area is reduced to make room for padding and border.

The initial value is `content-box` .

For example, start with an element in your XML that you want to size:

```
<notice>Keep this information with the procedure.</notice>
```

The selector `notice` matches that element name. With `content-box`, `inline-size` measures the content area. Padding and border are added outside it. With `border-box`, the specified size includes the padding and border:

```
notice {
    inline-size: 12em;
    padding: 1em;
    border: 1px solid black;
    box-sizing: content-box;
}
```

Here, the content area is `12em` wide. The padding adds `1em` on each inline side, and the border adds another `1px` on each side. Change `box-sizing` to `border-box` to keep the total border-edge size at `12em`; the content area becomes narrower to make room for the padding and border. The `padding` and `border` declarations make the edges visible in this example.

Preferred width, height, minimum and maximum dimensions are covered in Box dimensions.

Set margins outside the border

CSS Box Model 4 §3, Margins; CSS Logical Properties 1 §4.2, Flow-Relative Margins.

Physical margin properties

`margin-top` , `margin-right` , `margin-bottom` and `margin-left` set the space outside each physical side of the border.

Each accepts a length such as `12pt`, a percentage such as `5%`, `auto`, or `anchor-size()` . Negative lengths and percentages are allowed. A percentage is calculated from the inline size of the containing block, including for top and bottom margins. `auto` lets the layout algorithm distribute available space according to the formatting context.

`anchor-size()` accepts an optional anchor name, an optional size keyword, or both, followed by an optional length or percentage fallback. The size keyword can be `width`, `height`, `block`, `inline`, `self-block` or `self-inline`. The fallback is used when the anchor cannot be resolved.

Each physical margin longhand starts at `0`.

Adjacent block margins can collapse into one margin. Their used separation is not always the sum of both values.

`margin` shorthand

`margin` sets one to four physical side values. One value applies to all sides. Two values set top and bottom, then left and right. Three set top, left and right, then bottom. Four set top, right, bottom and left, clockwise from the top. Its initial value is `0`.

```
section { margin: 1em 2em; }
```

In this declaration, the section has `1em` of margin above and below it, and `2em` to its left and right. Replace `section` with the name of an element in your XML. The shorthand changes the space outside the border; it does not add space between the content and the border.

Logical margin properties

`margin-block-start` and `margin-block-end` set margins at the start and end of the block axis. `margin-inline-start` and `margin-inline-end` set margins at the start and end of the inline axis. Their sides follow `writing-mode` and `direction`.

`margin-block` sets the block-start and block-end margins. `margin-inline` sets the inline-start and inline-end margins. With one value both sides are equal; with two, the first sets the start side and the second the end side.

The four logical longhands accept the same values as the physical margin properties and start at `0`. Each logical shorthand accepts one or two values from that same grammar. The shorthand's initial values come from its longhands, each of which starts at `0`.

For example, a `step` element can have different spacing before and after it, and equal spacing at both inline sides:

```
<procedure>
    <step>Close the cover.</step>
    <step>Check the indicator.</step>
</procedure>
```

```
step {
    margin-block: 0.5em 1em;
    margin-inline: 1em;
}
```

`margin-block` gives each step `0.5em` at block-start and `1em` at block-end. `margin-inline` gives it `1em` at both inline sides. In the usual horizontal, left-to-right writing mode, these are top, bottom, left and right respectively. In another writing mode or direction, the logical sides follow that mode instead of staying tied to the same physical edges. Replace `step` with the element name used for your procedure steps.

`margin-trim`

`margin-trim` removes in-flow child margins that touch the corresponding edge of a container.

Its initial value is `none` .

- `none` does not trim margins. Normal margin collapsing can still occur.
- `block` trims block-start and block-end margins at the container's edges.
- `inline` trims inline-start and inline-end margins.
- `block-start` , `block-end` , `inline-start` and `inline-end` select individual edges.

You can combine `block` with `inline`, or combine individual edge keywords with one another. Do not mix these two groups in one value. The property applies to block, multi-column, flex and grid containers.

`margin-break`

`margin-break` controls the box's block-axis margins where they adjoin a page or column break. See CSS Fragmentation 4 §5.2.

- `auto` is the initial value. At an unforced break, margins adjoining the break are limited to the remaining space before the break and become zero after it. At a forced break, margins before the break are truncated, while margins after it are preserved. Cloned margins are always truncated to zero.

- `keep` preserves margins adjoining the break.
- `discard` truncates adjoining margins, including at the beginning and end of a fragmentation context.

When margins collapse, margins from more than one box can adjoin the same break.

For example, this rule removes a chapter's margins where they meet a page or column break:

```
chapter {
    margin-break: discard;
}
```

Set padding inside the border

CSS Box Model 4 §4, Padding; CSS Logical Properties 1 §4.4, Flow-Relative Padding.

Physical padding properties

`padding-top` , `padding-right` , `padding-bottom` and `padding-left` set the space between the content and the corresponding border edge.

Each accepts a non-negative length, such as `12pt`, or percentage, such as `5%`. A percentage is calculated from the inline size of the containing block, including for top and bottom padding. Padding cannot be negative.

Physical and logical padding longhands and shorthands have an initial value of `0`.

`padding` shorthand

`padding` sets one to four physical side values using the same top, right, bottom, left order as `margin`. One value applies to all sides; two set vertical then horizontal sides; three set top, horizontal sides, then bottom; four set each side clockwise.

For example, a `warning` element can have more room inside its border at the top and bottom than at the sides:

```
<warning>Disconnect the power before servicing.</warning>
```

```
warning {
    padding: 1em 1.5em;
}
```

The first value adds `1em` above and below the content. The second adds `1.5em` to the left and right. Padding is inside the border, unlike margin. Replace `warning` with the element name used for this kind of message in your XML.

Logical padding properties

`padding-block-start` and `padding-block-end` set padding at the start and end of the block axis. `padding-inline-start` and `padding-inline-end` set padding at the start and end of the inline axis.

`padding-block` sets both block-axis sides. `padding-inline` sets both inline-axis sides. One value applies to both sides; two values set the start side followed by the end side. The physical sides change with the writing mode and text direction.

Each logical padding longhand uses the same non-negative length-percentage values as the physical padding properties. Each logical shorthand accepts one or two values from the same grammar.

Use logical padding when the spacing should follow the text direction. In this example, `padding-inline-start` adds space at the beginning of the text line:

```
<definition>Meaning of the selected term.</definition>
```

```
definition {
    padding-block: 0.5em;
    padding-inline-start: 1em;
}
```

The block padding adds `0.5em` at both ends of the block axis. The inline-start padding adds `1em` before the text begins. In a right-to-left writing direction, inline-start is on the right. Replace `definition` with the element name used for this content in your XML.

Property reference

See the box model property reference for the generated value grammars, initial values, specification links, and implementation-check status.

Box dimensions

CSS sizing properties set preferred dimensions or place minimum and maximum limits on them. They apply to CSS boxes, not directly to XML tags. Select an element from your XML, then use the property that matches the dimension or limit you need.

For the meaning of CSS boxes and the length units used below, see Box sizing, margins and padding.

These properties also accept the CSS-wide keywords. See Inheritance and default values.

CSSMedia parses these declarations and computes their style values, but it does not lay out element boxes or generate a PDF. The properties therefore do not affect PDF output. The `Default` cells below and in the dimension property reference are left blank because there is no PDF output. The `Values` entries describe CSS syntax. CSS Anchor Positioning defines `anchor-size()` for physical sizing properties, not their logical counterparts.

Preferred sizes

CSS Sizing 3 §3, Specifying Box Sizes; CSS Logical Properties 1 §4.1, Logical Height and Logical Width.

`width` and `height` refer to the physical axes. `inline-size` and `block-size` refer to the text's inline and block axes. Their physical direction follows `writing-mode`.

Name	Description	Values	Default	Specification
`width`	Sets the preferred width of a box.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`height`	Sets the preferred height of a box.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`inline-size`	Sets the preferred size along the inline axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, or `calc-size()`.		CSS Sizing 3
`block-size`	Sets the preferred size along the block axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, or `calc-size()`.		CSS Sizing 3

- `<length-percentage>` means a length such as `12pt`, a percentage such as `60%`, or a `calc()` expression combining them.
- `[0,∞]` in `fit-content(<length-percentage [0,∞]>)` means the length or percentage cannot be negative.
- The `fit-content()` and `calc-size()` functions are defined in CSS Sizing 4 §3.2.
- `auto` lets the layout algorithm determine the size.
- `min-content` uses the smallest size the content can take without avoidable overflow.
- `max-content` uses the size the content would take without soft wrapping.
- `fit-content` fits the content within available space, bounded by its min-content and max-content sizes.
- `fit-content(<length-percentage>)` uses the supplied non-negative length or percentage as the available-space limit.
- `stretch` sizes the margin box to fill the available space in that axis.
- `contain` uses contain-fit sizing. It fills the available space while preserving a preferred aspect ratio when one exists.
- `calc-size()` calculates a size from an intrinsic-size basis. The basis can be an intrinsic keyword, another `calc-size()`, any, or a calculation. The second argument is a calculation using that basis.

- – `anchor-size()` accepts an optional anchor name, an optional size keyword, or both, followed by an optional length or percentage fallback. Its size keyword can be `width`, `height`, `block`, `inline`, `self-block` or `self-inline`. The function is for absolutely positioned boxes.
- A percentage size is relative to the corresponding dimension of the containing block. The allowed length and percentage forms are described in CSS Values 4 §5.5 and §6.

For example, if your XML contains a `chapter` element and a `figure` inside it, these rules request a 30em inline size for the chapter and a 60 percent inline size for the figure:

```
<chapter>
    <figure>Figure description and content.</figure>
</chapter>
```

```
chapter {
    inline-size: 30em;
}

figure {
    inline-size: 60%;
    block-size: fit-content;
}
```

Replace the example element names with names from your XML. If your document uses a vertical writing mode, inline and block dimensions map to different physical directions.

Minimum and maximum sizes

CSS Sizing 3 §3, Specifying Box Sizes; CSS Sizing 4 §3.1, Sizing Properties; CSS Logical Properties 1 §4.1.

Minimum sizes set a lower limit. Maximum sizes set an upper limit. The layout algorithm applies the limits to the preferred size.

Name	Description	Values	Default	Specification
`min-width`	Sets a lower limit for the width.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-height`	Sets a lower limit for the height.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

Name	Description	Values	Default	Specification
`min-inline-size`	Sets a lower limit along the inline axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, or `calc-size()`.		CSS Sizing 3
`min-block-size`	Sets a lower limit along the block axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, or `calc-size()`.		CSS Sizing 3
`max-width`	Sets an upper limit for the width.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-height`	Sets an upper limit for the height.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-inline-size`	Sets an upper limit along the inline axis.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, or `calc-size()`.		CSS Sizing 3
`max-block-size`	Sets an upper limit along the block axis.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, or `calc-size()`.		CSS Sizing 3

- `<length-percentage>` means a length such as `12pt`, a percentage such as `60%`, or a `calc()` expression combining them.
- `[0,∞]` in `fit-content(<length-percentage [0,∞]>)` means the length or percentage cannot be negative.
- `auto` on a minimum size lets the layout mode choose its automatic minimum. `none` on a maximum size means there is no maximum limit.
- For example, `max-inline-size` prevents a wide illustration from exceeding `40em`, while `inline-size: 100%` lets it use the containing block's full inline size when there is room:

```
<illustration>Diagram of the assembly.</illustration>
```

```
illustration {
    inline-size: 100%;
    max-inline-size: 40em;
}
```

Paired size shorthands

CSS Sizing 4 §3.1, Sizing Properties.

Name	Description	Values	Default	Specification
`size`	Sets both the preferred width and height.	One or two space-separated values. One value sets both dimensions; two set width then height.		CSS Sizing 4
`min-size`	Sets the minimum width and height.	One or two space-separated values. One value sets both dimensions; two set width then height.		CSS Sizing 4
`max-size`	Sets the maximum width and height.	One or two space-separated values. One value sets both dimensions; two set width then height.		CSS Sizing 4

- Each shorthand accepts one or two space-separated values. One value applies to both dimensions. Two values set width then height. Each value uses the grammar of its corresponding longhand.
- For `aspect-ratio` , the `contain-intrinsic-*` properties, `frame-sizing` , and `min-intrinsic-sizing` , see the complete dimension property reference.

Background color and images

A background can have a solid color, one or more images, or both. These properties describe the color and image layers of an element's background.

Procedure

1. `background-color`

   "CSS Backgrounds 4 §2.1, Base Color: the background-color property".

   Use `background-color` to set a solid color behind an element's content. Write a CSS color value, such as a named color, a hexadecimal color or an `rgb()` value.

   ```
   chapter {
       background-color: #f2f2f2;
   }
   ```

   The CSS initial value is `transparent`. In CSS, the background color is painted beneath the element's background images.

2. `background-image`

"CSS Backgrounds 4 §2.2, Image Sources: the background-image property".

Use `background-image` to specify images behind an element's content. A value can be a `url()` image, a gradient or none. Separate multiple layers with commas.

```
chapter {
    background-image: url("paper.png"), linear-gradient(white, #ddd);
}
```

The first image in the list is the top layer. Later images are below it, and the background color is below all image layers. Use none when a layer should not show an image.

Background positioning and size

These CSS properties specify where a background image is placed, how large it is and which box is used as its positioning area.

Procedure

1. `background-position`

   "CSS Backgrounds 4 §2.7, Background Positioning Shorthand: the background-position shorthand property".

   Use `background-position` to specify each background image's position inside its positioning area. A position can use keywords, lengths or percentages. With two values, the first sets the horizontal position and the second sets the vertical position.

   ```
   chapter {
       background-position: right 1em bottom 2em;
   }
   ```

   The CSS initial value is `0% 0%`, which places the image at the top-left. A percentage is calculated from the difference between the size of the positioning area and the size of the image. For example, `100% 100%` aligns the image's bottom-right corner with the area's bottom-right corner.

2. `background-position-x`, `background-position-y`, `background-position-inline` and `background-position-block`

   "CSS Backgrounds 4 §2.6, Background Positioning: the background-position-x, background-position-y, background-position-inline, and background-position-block properties".

   Use these properties to specify a position on one axis. `background-position-x` and `background-position-y` use the physical horizontal and vertical axes. `background-position-inline` and `background-position-block` use the logical axes, which follow the element's writing mode.

   ```
   chapter {
       background-position-x: right 1em;
       background-position-block: end 2em;
   }
   ```

   The physical properties use values such as `left`, `right`, `top` and `bottom`. The logical properties use `start` and end. Each also accepts `center`, a length or a percentage.

CSSMedia expands `background-position` into physical or logical axis values and also expands the `background` shorthand's position.

3. `background-size`

   "CSS Backgrounds 4 §2.10, Sizing Images: the background-size property".

   Use `background-size` to specify the width and height of each background image. Use one or two non-negative lengths or percentages. If you give one length or percentage, the other dimension defaults to `auto`.

   ```
   chapter {
       background-size: 12em auto;
   }
   ```

   `auto` uses the image's intrinsic size or proportion. `cover` scales the image until it covers the positioning area, possibly cropping part of it. `contain` scales it until the whole image fits, possibly leaving part of the area uncovered. Separate sizes for multiple image layers with commas.

4. `background-origin`

   "CSS Backgrounds 4 §2.9, Positioning Area: the background-origin property".

   Use `background-origin` to specify the box that defines the positioning area for background images.

   ```
   chapter {
       background-origin: content-box;
   }
   ```

   `border-box` uses the area inside the outer border edges, `padding-box` uses the area inside the outer padding edges, and `content-box` uses the content area. The CSS initial value is `padding-box`. This property does not affect an image whose `background-attachment` is `fixed`.

Background repetition and attachment

These CSS properties specify whether a background image repeats to fill its area and what the image is attached to.

Procedure

1. `background-repeat`

   "CSS Backgrounds 4 §2.4, Tiling Images Shorthand: the background-repeat property".

   Use `background-repeat` to specify whether an image repeats horizontally, vertically, both or neither. One value applies to both axes. Two values set the horizontal and vertical behavior, in that order.

```
chapter {
    background-repeat: no-repeat;
}

table {
    background-repeat: repeat no-repeat;
}
```

Use `repeat` to tile the image, `no-repeat` to show it once, `space` to distribute whole copies with space between them, or `round` to resize the copies so they fit evenly. `repeat-x` and `repeat-y` repeat on one physical axis only. `repeat-block` and `repeat-inline` repeat on the logical block or inline axis, respectively, and do not repeat on the other axis. Separate values for multiple image layers with commas.

CSSMedia expands `background-repeat` into its physical or logical axis properties.

2. `background-repeat-x`, `background-repeat-y`, `background-repeat-inline` and `background-repeat-block`

   "CSS Backgrounds 4 §2.3, Tiling Images: the background-repeat-x, background-repeat-y, background-repeat-block, and background-repeat-inline properties".

   Use these properties to specify repetition on one axis. The `x` and `y` properties use the physical horizontal and vertical axes. The `inline` and `block` properties use the logical axes, which follow the element's writing mode.

   ```
   chapter {
       background-repeat-x: repeat;
       background-repeat-block: no-repeat;
   }
   ```

   Each accepts `repeat`, `no-repeat`, `space` or `round`. Use the shorthand when you want to set both axes together.

3. `background-attachment`

   "CSS Backgrounds 4 §2.5, Affixing Images: the background-attachment property".

   Use `background-attachment` to specify how a background image is positioned in relation to its box or its contents. In paged media, CSS defines `fixed` relative to the page box, so that the background is repeated on each page.

   ```
   chapter {
       background-attachment: fixed;
   }
   ```

   In paged media, `fixed` attaches the image to the page box. CSSMedia's PDF default is `fixed`. Separate values for multiple image layers with commas.

Background clipping, shorthand and blending

These CSS properties specify the area where a background is painted, combine background settings in one declaration, and control how image layers combine.

Procedure

1. `background-clip`

   "CSS Backgrounds 4 §2.8, Painting Area: the background-clip property".

   Use `background-clip` to specify how far a background color or image extends across an element's box.

   ```
   chapter {
       background-clip: content-box;
   }
   ```

   `border-box` allows the background to extend beneath the border, `padding-box` clips it at the outside edge of the padding, and `content-box` clips it to the content area. `border-area` clips it to the area covered by the border. `text` clips it to the shape of the text glyphs. `border-area` and `text` can be combined. The CSS initial value is `border-box`. Separate values for multiple image layers with commas.

2. `background`

   "CSS Backgrounds 4 §2.12, Backgrounds Shorthand: the background property".

   Use `background` to specify several background properties in one declaration. For each layer, you can provide an image, position and size, repeat style, attachment and clipping or positioning boxes. Set the background color in the final layer.

   ```
   chapter {
       background: #f2f2f2 url("paper.png") center / cover no-repeat;
   }
   ```

   The slash separates `background-position` from `background-size`. A box value can set both the positioning and painting areas. Two box values set the positioning area first and the painting area second. The CSS shorthand resets its longhand properties to their initial values when you leave them out. It also resets `background-blend-mode` to `normal`.

3. `background-blend-mode`

   "Compositing and Blending 1 §3.4.3, The background-blend-mode property".

   Use `background-blend-mode` to specify how each background image layer combines with the layer below it and the element's background color.

   ```
   chapter {
       background-image: url("texture.png"), linear-gradient(white, gray);
       background-blend-mode: multiply, normal;
   }
   ```

   The available modes are `normal`, `multiply`, `screen`, `overlay`, `darken`, `lighten`, `color-dodge`, `color-burn`, `hard-light`, `soft-light`, `difference`, `exclusion`, hue, `saturation`, `color` and `luminosity`. The first value applies to the top image layer. If the list has fewer values than the image list, the values repeat. The CSS initial value is `normal`.

Shadows, opacity, transforms and masks

Multi-column layout

Flexbox and grid

Images, SVG and MathML

Tables

Cascade and declarations

CSS rules associate selectors with declarations. A declaration has a property name, a colon, a value and a semicolon:

```
chapter {
    color: navy;
    margin: 1em;
}
```

The **cascade** is the process that decides which declaration sets a property on an element when several declarations apply to it. It compares the declarations one criterion at a time and keeps the winner.

For each element and property, the cascade chooses among the declarations that apply. It does not choose one declaration for an entire rule. One rule can win for `color` while another wins for `margin`.

From a declaration to a value

CSS Cascade 5 §4, Value Processing; CSS Cascade 5 §5, Filtering.

CSS processes a property through several value stages:

1. A declared value is the value written in a declaration.
2. A cascaded value is the result of comparing the applicable declarations.
3. A specified value resolves CSS-wide keywords and uses inheritance or the property's initial value when needed.
4. A computed value resolves values that can be determined without knowing the final layout.
5. A used value incorporates information available during layout.
6. An actual value is the value that can be represented in the output medium.

The cascade concerns the first two stages. Inheritance and defaulting follow the cascade. Computation and layout follow those stages.

Declarations are filtered before they compete. A rule must apply to the element, its selector must match, and its enclosing conditions must hold. A declaration with an unknown property or invalid value does not provide a candidate. A value that becomes invalid only after substitution is different: it fails after the cascade has already chosen it. See Inheritance and default values and Custom properties and substitution.

CSSMedia processes declarations from its user-agent rules and author stylesheets. The cascade guides explain the CSS rules used to select values from those declarations.

Guides in this topic

- Cascade order and layers: importance, origins, layers, specificity, scope proximity and source order.
- Inheritance and default values: inherited values, initial values and CSS-wide keywords.
- Shorthand properties: how shorthand declarations set and reset longhands.
- Custom properties and substitution: custom properties, `var()`, `attr()` and `@property`.

- Conditional rules: when rules participate in the cascade and how imported stylesheets are ordered.

Cascade order and layers

When declarations set the same property on the same element, CSS sorts them by precedence. It compares one criterion at a time. The first criterion that differs decides which declaration wins.

Sorting declarations

CSS Cascade 6 §2.1, Cascade Sorting Order; CSS Cascade 5 §6.1, Cascade Sorting Order; Selectors 4 §15, Calculating a selector's specificity.

The cascade compares declarations in this order:

1. Origin and importance.
2. Encapsulation context.
3. Element-attached style declarations.
4. Cascade layer order.
5. Selector specificity.
6. Scope proximity.
7. Source order.

Each step matters only when the declarations tie at every preceding step.

Origin and importance

CSS Cascade 5 §6.1, Cascade Sorting Order.

An origin identifies the source of a declaration. CSS defines user-agent, user, and author origins, plus animation and transition origins. Within the author, user, and user-agent origins, important declarations and normal declarations have different priorities. From highest to lowest, the CSS cascade order is:

Priority	Declaration
1	Transition
2	Important user-agent
3	Important user
4	Important author
5	Animation
6	Normal author
7	Normal user
8	Normal user-agent

- `!important` follows the value and raises that declaration above normal declarations in its origin. It does not make a declaration universally unbeatable. A higher-priority origin can still win.

- CSSMedia's stylesheet input consists of author stylesheets and built-in user-agent rules. It does not load a user-origin stylesheet.

Encapsulation context

CSS Cascade 6 §2.1, Cascade Sorting Order.

Some host environments divide a document into nested encapsulation contexts, such as shadow trees. For normal declarations, the outer context wins a tie against an inner context. For important declarations, the inner context wins. This comparison occurs after origin and importance, and before layers.

Element-attached styles

CSS Cascade 6 §2.1, Cascade Sorting Order; CSS Style Attributes §2, CSS Style Attributes.

When a host language defines a style attribute, declarations attached directly to an element take precedence over declarations from style rules at the same origin and importance. This rule is separate from selector specificity.

Cascade layers

CSS Cascade 5 §6.4, Cascade Layers; CSS Cascade 5 §6.4.3, Layer Ordering.

`@layer` groups declarations within an origin. A layer's order is established when its name first appears. A layer statement establishes order without adding rules. A layer block establishes a layer and contains its rules. Layers can be nested; a dotted name such as `components.buttons` identifies a sublayer.

For normal declarations, later layers outrank earlier layers. Unlayered declarations form an implicit final layer and outrank normal declarations in every named layer. For important declarations, this order is reversed: earlier layers outrank later layers, and layered important declarations outrank unlayered important declarations.

Within a parent layer, declarations assigned directly to the parent are ordered after its sublayers for normal declarations. Important declarations reverse that order as well.

```
@layer base, theme;

@layer base {
    chapter { color: black; }
}

@layer theme {
    chapter { color: navy; }
}

chapter { color: gray; }
```

All three declarations are normal and have the same specificity. The unlayered `gray` declaration wins. If all three declarations were important, the `base` layer would win.

Specificity

Selectors 4 §15, Calculating a selector's specificity.

Specificity compares the selector that matched. It is a three-part weight: ID selectors, class/attribute/pseudo-class selectors, and type/pseudo-element selectors. Compare the parts from left to right. A larger earlier part takes precedence, regardless of the later parts.

For a selector list, the matching selector with the greatest specificity supplies the rule's specificity. Specificity does not override origin, importance or layer order. See Specificity for how each selector form contributes.

Scope proximity

CSS Cascade 6 §2.5, Scoping Styles: the @scope rule.

When declarations tie on origin, importance, layer and specificity, a declaration from a scope whose root is closer to the matched element wins. Count the ancestor steps from the element to each matching scope root. The smaller count is closer. An unscoped rule has no scope root and loses this tie-breaker to a scoped rule.

Source order

CSS Cascade 6 §2.1, Cascade Sorting Order.

If declarations still tie, the one that appears later wins. Imported rules take part at the position of their `@import`. Within one declaration block, the later declaration for the same property wins.

Scoped rules

CSS Cascade 6 §2.5, Scoping Styles: the @scope rule.

`@scope` limits the elements matched by its nested rules. The optional start selector identifies the scope root. The optional `to` selector identifies a scope limit: matching elements at the limit, and their descendants, are outside that scope.

```
@scope (chapter) to (appendix) {
    title { color: navy; }
}
```

This rule matches `title` elements inside a `chapter` scope, but not inside an `appendix` boundary. If the start selector is omitted, the document root is the scope root. A nested scope is relative to its enclosing scope.

Scope proximity affects precedence only after origin, importance, layer and specificity tie. It does not make a less-specific selector outrank a more-specific one.

Page-context declarations

CSS Paged Media 3 §4.4, Cascading in the page context; CSS GCPM 3 §3.1, Page Selectors; §3.2, Page groups.

Declarations in @page rules and page-margin rules cascade within their page context. Page selectors have their own specificity. A page type name contributes to the first component; `:first` and `:blank` contribute to the second; `:left` and `:right` contribute to the third. Each occurrence counts, including repeated pseudo-classes. The `:nth()` page pseudo-class contributes no specificity. It selects a page by its position in the document or, with a page-group name, its position in that group. When matching page rules tie on origin, importance, layer and specificity, source order decides. These declarations do not compete with declarations for an XML element.

Inheritance and default values

The cascade selects a cascaded value for a property on an element. CSS then determines the property's specified value. If no declaration supplies a cascaded value, inheritance or the property's initial value provides the default.

Inheritance

CSS Cascade 5 §7.2, Inheritance.

Some properties inherit automatically. Their definitions say whether they are inherited. A child element uses the parent's computed value unless a declaration gives the child another value.

```
chapter { color: navy; }
title { color: teal; }
```

The title uses teal. Other descendants inherit navy. A property that does not normally inherit uses its initial value when it has no cascaded value.

Inheritance follows the XML element tree. It is not determined by whether the property describes text, a box, or another feature.

CSS-wide keywords

CSS Cascade 5 §7.3, Explicit Defaulting.

CSS-wide keywords can be used as the complete value of any property. They affect defaulting after the cascade has selected the declaration.

`initial`

CSS Cascade 5 §7.3.1, Resetting a Property: the initial keyword.

`initial` selects the property's initial value from its definition. It does not mean the value inherited from the parent, even for a property that normally inherits.

```
title { color: initial; }
```

`inherit`

CSS Cascade 5 §7.3.2, Explicit Inheritance: the inherit keyword.

`inherit` uses the parent's computed value, whether or not the property normally inherits. At the root, where there is no parent, it uses the initial value.

```
title { margin-left: inherit; }
```

`unset`

CSS Cascade 5 §7.3.3, Erasing All Declarations: the unset keyword.

`unset` acts like `inherit` for a property that normally inherits. It acts like `initial` for a property that does not normally inherit.

```
chapter { color: unset; }
title { margin-left: unset; }
```

The first declaration follows the normal inheritance behavior of `color`. The second uses the initial value of `margin-left`.

`revert`

CSS Cascade 5 §7.3.4, Rolling Back Cascade Origins: the revert keyword.

`revert` removes declarations from the winning declaration's origin for that property and runs the cascade again. For an author declaration, this means the author-origin value is rolled back so a lower origin can supply a value. If no declaration remains, normal defaulting and inheritance apply.

```
title { color: navy; }
title { color: revert; }
```

`revert-layer`

CSS Cascade 5 §7.3.5, Rolling Back Cascade Layers: the revert-layer keyword.

`revert-layer` removes declarations from the winning declaration's layer at that origin and reruns the cascade. A declaration in a lower-priority layer can then win. If no declaration remains in that origin, the cascade can fall back to a lower origin.

```
@layer base, theme;

@layer base {
    title { color: black; }
}

@layer theme {
    title { color: revert-layer; }
}
```

The `theme` declaration rolls back to the `base` layer's value.

The `all` property

CSS Cascade 5 §3.2, Resetting All Properties: the all property.

The `all` property accepts a CSS-wide keyword and applies it to nearly every property. It does not affect `direction`, `unicode-bidi`, or custom properties.

```
title { all: unset; }
```

This uses each property's normal `unset` behavior. Inherited properties inherit; other properties use their initial values. Use `all` carefully because it can reset declarations from many unrelated property groups.

A declaration invalid after the cascade

CSS Variables 1 §2.3, Resolving Dependency Cycles; CSS Variables 1 §3.1, Invalid Variables.

A value can pass the declaration's initial syntax check and become invalid later, for example when a `var()` reference has no usable value or substitution produces text that does not match the property's grammar. The cascade has already selected that declaration. CSS does not return to an earlier, lower-priority declaration for that property; the invalid-at-computed-value-time declaration acts as `unset`.

This differs from a declaration that is invalid before it enters the cascade. An invalid declaration is discarded, so another valid declaration can win.

Shorthand properties

A shorthand sets several related longhand properties in one declaration. The longhands are the properties that hold the individual values.

Shorthands set every sub-property

CSS Cascade 5 §3, Shorthand Properties.

A shorthand sets all of its sub-properties, including properties whose values are omitted from the shorthand. Unless the shorthand's definition says otherwise, each omitted sub-property receives its initial value.

For example, `margin` sets the four side margins:

```
chapter { margin: 1em; }
chapter { margin: 1em 2em; }
chapter { margin: 1em 2em 3em 4em; }
```

One value sets all four sides. Two values set the top and bottom margins, then the right and left margins. Three values set top, horizontal sides and bottom. Four values set top, right, bottom and left, respectively.

Some shorthands also reset sub-properties that do not appear in their syntax. Such a sub-property is reset-only. For example, the `border` shorthand resets border-image properties even though its syntax cannot set their other values. Consult each shorthand's property definition to see its full set of sub-properties and reset behavior.

Shorthand values and importance

CSS Cascade 5 §3, Shorthand Properties.

The shorthand is expanded into its longhands at the location of its declaration. Those longhands then compete in the cascade like declarations written separately.

```
chapter {
    margin-left: 2em;
    margin: 1em;
}
```

The `margin` shorthand comes later and sets `margin-left` to `1em`. If a later declaration sets `margin-left` directly, that later longhand can win.

An invalid shorthand contributes no longhand declarations. If a shorthand is declared `!important`, its sub-properties are important too. A CSS-wide keyword used as the shorthand's entire value applies to every sub-property, including reset-only sub-properties. A CSS-wide keyword cannot be combined with other shorthand values.

Resetting properties with `all`

CSS Cascade 5 §3.2, Resetting All Properties: the all property.

`all` is a shorthand that sets almost every property to one CSS-wide keyword. It excludes `direction`, `unicode-bidi` and custom properties.

```
title { all: initial; }
```

This sets the included properties to their initial values. Use `inherit`, `unset`, `revert` or `revert-layer` to apply the corresponding CSS-wide behavior to those properties. See Inheritance and default values for the meaning of each keyword.

Shorthand definitions

Shorthands do not all use the same value order or number of values. Their property definitions specify the grammar, defaults for omitted values and reset-only sub-properties. The guides for backgrounds, fonts and other property groups describe the shorthands in those groups.

Custom properties and substitution

Custom properties store values that can be substituted into other declarations. The cascade selects their values like it selects values for other properties.

Procedure

1. Defining custom properties

   CSS Variables 1 §2, Defining Custom Properties: the --* family of properties.

   A custom property name is a dashed identifier that begins with two hyphens, such as `--accent`. Names are case-sensitive. An unregistered custom property accepts a sequence of CSS component values rather than a property's normal value grammar.

   ```
   chapter {
       --accent: navy;
       --space: 1em;
   }
   ```

   Custom properties participate in the cascade. Unregistered custom properties inherit by default, and their initial value is the guaranteed-invalid value. This value is different from an empty string and cannot be substituted unless a fallback is provided.

2. Substituting values with `var()`

   CSS Variables 1 §3, Using Cascading Variables: the var() notation; CSS Variables 1 §2.3, Resolving Dependency Cycles; CSS Variables 1 §3.1, Invalid Variables.

   Use `var()` with a custom property name. An optional fallback follows a comma:

   ```
   title {
       color: var(--accent, black);
       margin-left: var(--space, 0);
   }
   ```

   The fallback is used when the referenced custom property has the guaranteed-invalid value. A fallback can contain another `var()` call. The fallback is not used merely because the substituted text fails the consuming property's grammar.

Custom properties can refer to one another. CSS resolves their dependencies for each element. A cycle makes every custom property in that cycle invalid on that element. Descendants inherit the parent's computed custom-property values; they do not re-evaluate the parent's declarations in the descendant's context.

Substitution happens after the cascade. If the resulting value does not match the consuming property's grammar, the winning declaration becomes invalid at computed-value time and acts as `unset`. The cascade does not select an earlier declaration instead.

3. Registering a custom property with `@property`

   CSS Properties and Values API 1 §3, The @property Rule; §3.1, The syntax Descriptor; §3.2, The inherits Descriptor; §3.3, The initial-value Descriptor.

   `@property` registers a custom property with a syntax, an inheritance setting and, for typed syntax, an initial value:

   ```
   @property --accent {
       syntax: "<color>";
       inherits: true;
       initial-value: navy;
   }
   ```

   The `syntax` descriptor is a string containing the property's accepted grammar. `"*"` accepts an arbitrary token sequence. A typed syntax uses component types such as `"<length>"`, which accepts a length such as `12pt`, or `"<color>"`, which accepts a CSS color. It can combine alternatives with `|` and use `+` or `#` multipliers. The `inherits` descriptor is required and must be `true` or `false`. A typed syntax requires an `initial-value` that matches the syntax and is computationally independent. With `"*"`, `initial-value` may be omitted.

   A registered property's value is checked against its syntax when computed. Its `inherits` descriptor controls whether it inherits when no value is specified. Its `initial-value` supplies the default when it does not inherit or has no inherited value. If more than one valid registration names the same custom property, the last one in stylesheet order is used.

4. Reading XML attributes with `attr()`

   CSS Values 5 §7.7, Attribute References: the attr() notation.

   `attr()` reads an attribute from the XML element being styled. Without a type, it produces a string. A syntax can request a parsed CSS value, and an optional fallback supplies a value when the attribute is absent or cannot be parsed:

   ```
   figure {
       content: attr(title);
       width: attr(data-width <length>, 100pt);
   }
   ```

   In `attr(data-width <length>, 100pt)`, `<length>` means a CSS length such as `12pt`. An untyped `attr()` has an empty-string fallback when none is written. A typed `attr()` has no implicit fallback. Attribute references can include a namespace prefix declared by `@namespace`.

An attribute-derived value is tainted. If it is used as or within a URL, the declaration becomes invalid at computed-value time. After substitution, the resulting declaration must still match the property's grammar.

Conditional rules

A conditional rule contains rules that participate in the cascade only when its condition is true. A condition filters rules; it does not give declarations extra priority. Declarations that pass the condition compete by the usual cascade order.

Procedure

1. Media conditions with `@media`

   CSS Conditional 3 §5, Media-specific style sheets: the @media rule; Media Queries 5 §2, Media Queries.

   `@media` tests a media query list. Queries can test a media type, media features, or both. A comma separates alternatives: the list matches if at least one query is true. `and` combines a media type or feature with another test. `not` negates the complete query. `only` has no effect on whether a query matches.

   ```
   @media print and (width >= 6in) {
       chapter { color: black; }
   }
   ```

   CSSMedia evaluates media types for paged output. `print` and `all` match. `screen` does not. A query without a media type is treated as `all`. Media feature queries are evaluated against the page and environment values CSSMedia uses.

2. Feature tests with `@supports`

   CSS Conditional 3 §6, Feature queries: the @supports rule; CSS Conditional 5 §2, Extensions to the @supports rule.

   `@supports` tests whether CSSMedia accepts a declaration or selector. A declaration test uses parentheses:

   ```
   @supports (color: navy) {
       title { color: navy; }
   }
   ```

   Use `selector()` to test one complex selector. CSS also defines `font-tech()` and `font-format()` tests. `at-rule()` tests whether an at-rule is supported, and a named support condition can test a group declared with `@supports-condition`. Conditions can combine tests with `not`, `and` and `or`; use parentheses when combining `and` and `or`.

   ```
   @supports selector(chapter > title) and (display: block) {
       chapter > title { display: block; }
   }
   ```

   A true feature test means the tested syntax is accepted. It does not test whether a selector matches any element or whether a particular document contains a value.

3. Named support conditions

CSS Conditional 5 §8, Defining Custom Support Queries: the @supports-condition rule.

`@supports-condition` gives a name to a group of feature tests. Declarations and nested selectors in the block are tested for support; they do not add styling declarations. Define the name before using it:

```
@supports-condition --basic-layout {
    display: block;
    color: navy;
}

@supports (--basic-layout) {
    chapter { display: block; }
}
```

If more than one definition uses the same name, the last definition in document order replaces the earlier one.

4. General conditions with `@when`

   CSS Conditional 5 §3, Generalized Conditional Rules: the @when rule.

   `@when` tests a boolean condition that can combine `media()` and `supports()` tests. It provides a single condition syntax for tests that would otherwise appear in separate `@media` or `@supports` rules.

   ```
   @when media(overflow-block: paged) and supports(color: navy) {
       title { color: navy; }
   }
   ```

   Conditions can be grouped with parentheses and combined with `and`, `or` and `not`.

5. Conditional chains with `@else`

   CSS Conditional 5 §4, Chained Conditionals: the @else rule.

   An `@else` rule follows an `@when`, `@media`, `@supports` or `@container` rule to form a chain. Only the first branch whose condition is true applies. `@else` can have its own condition; without one, it is the final fallback.

   ```
   @when supports(color: navy) {
       title { color: navy; }
   } @else supports(color: black) {
       title { color: black; }
   } @else {
       title { color: inherit; }
   }
   ```

   Each `@else` belongs to the immediately preceding conditional chain in the same rule list. An unrelated rule or declaration ends the chain.

6. Reusable media queries with `@custom-media`

   Media Queries 5 §10, Custom Media Queries.

   `@custom-media` assigns a name to a media query list, or to the boolean value `true` or `false`. Use the name in a media query by writing it inside parentheses:

```
@custom-media --wide-page (width >= 6in);

@media (--wide-page) {
    chapter { color: black; }
}
```

A definition can refer to other custom media names. Circular references do not produce a true query. When a name is defined more than once, the last definition is used.

7. Container queries with `@container`

   CSS Conditional 5 §5.1, Creating Query Containers: the container-type property; §5.4, Container Queries: the @container rule.

   `@container` selects a query container from the element's ancestors. A container name before the condition restricts which ancestor is selected. A style query tests a computed style or custom property on the selected container. Conditions can combine tests with `and`, `or` and `not`.

   ```
   @container style(--theme: dark) {
       chapter { color: white; }
   }
   ```

   A style query can use an element as a container without size containment. Without a name, the nearest eligible ancestor is used. With a name, the nearest ancestor whose `container-name` includes that name is used.

8. Importing stylesheets

   CSS Cascade 5 §2, Importing Style Sheets: the @import rule; CSS Cascade 5 §2.1, Conditional @import Rules.

   `@import` loads another stylesheet. Its rules take part in the cascade at the import's position. The URL or quoted string comes first. Optional clauses can assign the imported rules to a layer, test feature support, or restrict them to matching media:

   ```
   @import "print.css" layer(print-theme) supports(color: navy) print;
   ```

   An optional `@charset`, layer-order statements and `@supports-condition` rules can precede imports. Imports precede `@namespace`; namespace declarations precede ordinary rules. An import after another rule is ignored. Imports can be conditional. If an import condition is false, its rules do not participate. A layer clause places the imported stylesheet in that layer; an unnamed `layer` creates an anonymous layer.

9. Layer rules

   CSS Cascade 5 §6.4, Cascade Layers.

   `@layer` declares a layer order or groups rules into a layer:

   ```
   @layer reset, base, theme;

   @layer theme {
       title { color: navy; }
   }
   ```

Declarations in layers use the layer precedence described in Cascade order and layers. Layer statements establish order, and layer blocks group rules. An `@import` can assign its imported rules to a layer with a layer clause.

10. Other stylesheet declarations

    CSS Namespaces 3 §3.1, Syntax.

    `@namespace` declares prefixes used by selectors. See Basic selectors and namespaces. Conditional rules and imports can also contain @page rules; page declarations use the page-context cascade described in Cascade order and layers.

Reference

Look up properties, values and at-rules by subject.

Each table links to the CSS specification that defines its entries. The references are grouped by subject:

- Selector reference
- Font reference
- Box model properties
- Dimension properties
- Background and image reference
- Display and positioning
- Page reference
- Links and generated content reference
- Cascade reference

Selector reference

This table will list the selectors and rules covered in this chapter.

Name	Description	Values	Default	Specification

Font reference

This table covers the font properties, at-rules, and descriptors introduced in the font guides.

Name	Description	Values	Default	Specification
`font`	Sets the font family, size, style, weight, width, variant, and line height in one declaration.	A system-font keyword (`caption`, `icon`, `menu`, `message-box`, `small-caption`, or `status-bar`), or optional font-style, `normal` or `small-caps`, font-weight, and font-width values followed by required font size and family and optional line height		CSS Fonts 4, `font`

Name	Description	Values	Default	Specification
`font-family`	Selects a font family or ordered list of fallback families.	Family names and `serif`, `sans-serif`, `system-ui`, `cursive`, `fantasy`, `math`, `monospace`, `ui-serif`, `ui-sans-serif`, `ui-monospace`, `ui-rounded`, or `generic(fangsong)`, `generic(kai)`, `generic(khmer-mul)`, `generic(nastaliq)`		CSS Fonts 4, `font-family`
`font-size`	Sets the font size.	`xx-small`, `x-small`, `small`, `medium`, `large`, `x-large`, `xx-large`, `xxx-large`, `larger`, `smaller`, a non-negative length or percentage, or `math`		CSS Fonts 4, `font-size`
`font-size-adjust`	Adjusts font sizing using a selected font metric.	none, optional metric (`ex-height`, `cap-height`, `ch-width`, `ic-width`, or `ic-height`), then `from-font` or a non-negative number		CSS Fonts 5, `font-size-adjust`
`font-weight`	Selects the requested weight of a font face.	`normal`, `bold`, `bolder`, `lighter`, or a number from 1 to 1000		CSS Fonts 4, `font-weight`
`font-width`	Selects the requested width of a font face.	`normal`, a non-negative percentage, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded`, or `ultra-expanded`		CSS Fonts 4, `font-width`
`font-stretch`	Legacy alias for `font-width`.	`normal`, a non-negative percentage, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded`, or `ultra-expanded`		CSS Fonts 4, `font-stretch`

Name	Description	Values	Default	Specification
`font-style`	Selects the requested upright or slanted face.	`normal`, `italic`, `left`, `right`, or `oblique` with an optional angle from -90deg to 90deg		CSS Fonts 4, `font-style`
`font-synthesis`	Controls synthetic weight, style, small-cap, and position forms.	none, or any combination of `weight`, `style`, `small-caps`, and `position`		CSS Fonts 4, `font-synthesis`
`font-synthesis-weight`	Controls synthetic bolding.	`auto` or none		CSS Fonts 4, `font-synthesis-weight`
`font-synthesis-style`	Controls synthetic italic or oblique styling.	`auto`, none, or `oblique-only`		CSS Fonts 4, `font-synthesis-style`
`font-synthesis-small-caps`	Controls synthetic small capitals.	`auto` or none		CSS Fonts 4, `font-synthesis-small-caps`
`font-synthesis-position`	Controls synthetic subscript and superscript forms.	`auto` or none		CSS Fonts 4, `font-synthesis-position`
`font-variant`	Shorthand for typographic variant properties.	`normal`, none, or combinations of variant longhand values		CSS Fonts 4, `font-variant`
`font-variant-alternates`	Requests named alternate glyphs and stylistic forms.	`normal` or combinations of `stylistic()`, `historical-forms`, `styleset()`, `character-variant()`, `swash()`, `ornaments()`, and `annotation()`		CSS Fonts 4, `font-variant-alternates`

Name	Description	Values	Default	Specification
font-variant-caps	Requests small-cap and other capitalization forms.	normal, small-caps, all-small-caps, petite-caps, all-petite-caps, unicase, or titling-caps		CSS Fonts 4, font-variant-caps
font-variant-east-asian	Requests East Asian glyph variants and widths.	normal or combinations of jis78, jis83, jis90, jis04, simplified, traditional, full-width, proportional-width, and ruby		CSS Fonts 4, font-variant-east-asian
font-variant-emoji	Requests text or emoji presentation for supported characters.	normal, text, emoji, or unicode		CSS Fonts 4, font-variant-emoji
font-variant-ligatures	Controls common, discretionary, historical, and contextual ligatures.	normal, none, or combinations of common-ligatures/no-common-ligatures, discretionary-ligatures/no-discretionary-ligatures, historical-ligatures/no-historical-ligatures, and contextual/no-contextual		CSS Fonts 4, font-variant-ligatures
font-variant-numeric	Requests number shapes, spacing, and fractions.	normal or combinations of lining-nums/oldstyle-nums, proportional-nums/tabular-nums, diagonal-fractions/stacked-fractions, ordinal, and slashed-zero		CSS Fonts 4, font-variant-numeric
font-variant-position	Requests subscript or superscript glyph forms.	normal, sub, or super		CSS Fonts 4, font-variant-position

Name	Description	Values	Default	Specification
`font-kerning`	Controls use of kerning information in the font.	`auto`, `normal`, or none		CSS Fonts 4, `font-kerning`
`font-feature-settings`	Sets low-level OpenType feature tags and values.	`normal` or a comma-separated list of feature tags, optionally followed by a non-negative integer, on, or `off`		CSS Fonts 4, `font-feature-settings`
`font-language-override`	Sets the OpenType language system tag.	`normal` or a string		CSS Fonts 4, `font-language-override`
`font-optical-sizing`	Enables or disables automatic optical-size axis selection for variable fonts.	`auto` or none		CSS Fonts 4, `font-optical-sizing`
`font-variation-settings`	Sets low-level variable-font axis tags and values.	`normal` or a comma-separated list of four-character tags and numbers		CSS Fonts 4, `font-variation-settings`
`font-palette`	Chooses a color palette for a color font.	`normal`, `light`, `dark`, a palette identifier, or `palette-mix()`		CSS Fonts 4, `font-palette`
`line-height`	Sets the preferred height of inline content used to determine line-box height.	`normal`, a non-negative number, non-negative length, or non-negative percentage		CSS Inline 3, `line-height`
`@font-face`	Declares a font face and its source for use by a family name.	Font-face rule with descriptors		CSS Fonts 4, `@font-face`

Name	Description	Values	Default	Specification
`@font-face font-family`	Names the family for this face.	A family name		CSS Fonts 4, `font-family` descriptor
`@font-face src`	Lists font resources or installed font names to try in order.	Comma-separated `url()` or `local()` sources with optional format and technology hints		CSS Fonts 4, `src` descriptor
`@font-face font-display`	Sets the fallback and swap behavior while a font loads.	`auto`, `block`, `swap`, `fallback`, or `optional`		CSS Fonts 4, `font-display` descriptor
`@font-face font-weight`	Describes the weight or weight range supplied by the face.	`auto`, or one or two values, each `normal`, `bold`, or a number from 1 to 1000		CSS Fonts 4, `font-weight` descriptor
`@font-face font-width`	Describes the width or width range supplied by the face.	`auto`, or one or two `font-width` values, each `normal`, a width keyword, or a non-negative percentage		CSS Fonts 4, `font-width` descriptor
`@font-face font-stretch`	Legacy alias for the `font-width` descriptor.	`auto`, or one or two values, each `normal`, a non-negative percentage, `ultra-condensed`, `extra-condensed`, `condensed`, `semi-condensed`, `semi-expanded`, `expanded`, `extra-expanded`, or `ultra-expanded`		CSS Fonts 4, `font-stretch` descriptor
`@font-face font-style`	Describes the style or style range supplied by the face.	`auto`, `normal`, `italic`, `left`, `right`, or `oblique` with no angle or one or two angles from -90deg to 90deg		CSS Fonts 4, `font-style` descriptor
`@font-face unicode-range`	Limits which character code points this face can serve.	One or more Unicode ranges		CSS Fonts 4, `unicode-range` descriptor

Name	Description	Values	Default	Specification
`@font-face font-feature-settings`	Sets default low-level OpenType feature settings for a face.	`normal` or a comma-separated list of feature tags, optionally followed by a non-negative integer, on, or `off`		CSS Fonts 4, `font-feature-settings` descriptor
`@font-face font-language-override`	Sets the default OpenType language system tag for a face.	`normal` or a string		CSS Fonts 4, `font-language-override` descriptor
`@font-face font-named-instance`	Selects a named variation instance in the font.	`auto` or a string		CSS Fonts 4, `font-named-instance` descriptor
`@font-face font-variation-settings`	Sets the face's default variable-font axes.	`normal` or a list of axis tags and numbers		CSS Fonts 4, `font-variation-settings` descriptor
`@font-face font-size`	Describes the face's optical-size value or range.	`auto` or one or two numbers		CSS Fonts 5, `font-size` descriptor
`@font-face ascent-override`	Overrides the face's ascent metric.	One or two values, each `normal` or a non-negative percentage		CSS Fonts 5, `ascent-override`
`@font-face descent-override`	Overrides the face's descent metric.	One or two values, each `normal` or a non-negative percentage		CSS Fonts 5, `descent-override`
`@font-face line-gap-override`	Overrides the face's line-gap metric.	One or two values, each `normal` or a non-negative percentage		CSS Fonts 5, `line-gap-override`
`@font-face size-adjust`	Scales the face's glyph outlines and metrics.	Non-negative percentage		CSS Fonts 5, `size-adjust`

Name	Description	Values	Default	Specification
`@font-face subscript-position-override`	Overrides the subscript position metric.	One or two values, each `normal`, `from-font`, or a percentage		CSS Fonts 5, `subscript-position-override`
`@font-face subscript-size-override`	Overrides the subscript size metric.	One or two values, each `normal`, `from-font`, or a non-negative percentage		CSS Fonts 5, `subscript-size-override`
`@font-face superscript-position-override`	Overrides the superscript position metric.	One or two values, each `normal`, `from-font`, or a percentage		CSS Fonts 5, `superscript-position-override`
`@font-face superscript-size-override`	Overrides the superscript size metric.	One or two values, each `normal`, `from-font`, or a non-negative percentage		CSS Fonts 5, `superscript-size-override`
`@font-feature-values`	Names feature values for one or more font families.	Rule with one or more family names and feature-value blocks		CSS Fonts 4, `@font-feature-values`
`@font-feature-values font-display`	Sets a default font-display policy for the listed families.	`auto`, `block`, `swap`, `fallback`, or `optional`		CSS Fonts 4, `font-display` descriptor
`@font-feature-values @annotation`	Maps names to annotation feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@annotation`
`@font-feature-values @character-variant`	Maps names to character-variant feature indexes.	One or two integers. Values 1 to 99 select `cv01` to `cv99`; in a two-integer value, the second integer sets that feature's value. Zero and values above 99 parse but select no feature.		CSS Fonts 4, `@character-variant`

Name	Description	Values	Default	Specification
`@font-feature-values @historical-forms`	Maps names to historical-forms feature indexes.	Named values mapped to one or more non-negative integers		CSS Fonts 4, `@historical-forms`
`@font-feature-values @ornaments`	Maps names to ornaments feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@ornaments`
`@font-feature-values @styleset`	Maps names to styleset feature indexes.	Named values mapped to one or more integers; 1 through 99 select `ss01` through `ss99`, while 0 and values above 99 select no OpenType feature		CSS Fonts 4, `@styleset`
`@font-feature-values @stylistic`	Maps names to stylistic feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@stylistic`
`@font-feature-values @swash`	Maps names to swash feature indexes.	Named values mapped to one non-negative integer		CSS Fonts 4, `@swash`
`@font-palette-values`	Defines a named palette for a color font.	Rule with a dashed identifier and palette descriptors		CSS Fonts 4, `@font-palette-values`
`@font-palette-values font-family`	Required descriptor that identifies the font family to which the named palette applies.	One or more family names		CSS Fonts 4, `font-family` descriptor
`@font-palette-values base-palette`	Chooses the base palette for the named palette.	`light`, `dark`, or a non-negative integer		CSS Fonts 4, `base-palette`

Name	Description	Values	Default	Specification
`@font-palette-values override-colors`	Replaces selected colors in the base palette.	One or more palette index and color pairs		CSS Fonts 4, `override-colors`

Box model properties

This table lists the box model properties described in Box sizing, margins and padding. A length is a number with a unit, such as `12pt` or `1em`. Margin and padding percentages are relative to the containing block's logical width, or inline size.

CSSMedia accepts and computes these declarations, but does not use element box model values to lay out element boxes. PDF generation is not implemented. Page geometry is prepared separately. The `Default` cells are blank because there is no PDF output.

The CSS-wide keywords `inherit`, `initial`, `revert`, `revert-layer`, and `unset` can also be used with these properties.

Name	Description	Values	Default	Specification
`box-sizing`	Chooses whether `width` and `height` measure the content area or the box including padding and border.	`content-box` measures the content; `border-box` includes padding and border.		CSS Sizing 3
`margin`	Sets the space outside all four sides of a box.	One to four lengths or percentages, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-block`	Sets the margins at the start and end of the block axis.	One or two lengths, percentages, `auto`, or `anchor-size()` values.		CSS Logical Properties 1
`margin-block-end`	Sets the margin at the end of the block axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-block-start`	Sets the margin at the start of the block axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-bottom`	Sets the space outside the bottom side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4

Name	Description	Values	Default	Specification
`margin-break`	Controls whether a margin is kept or discarded when its box is split by a page break.	`auto` follows the break rules; `keep` keeps the margin; `discard` removes it.		CSS Fragmentation 4
`margin-inline`	Sets the margins at the start and end of the inline axis.	One or two lengths, percentages, `auto`, or `anchor-size()` values.		CSS Logical Properties 1
`margin-inline-end`	Sets the margin at the end of the inline axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-inline-start`	Sets the margin at the start of the inline axis.	Length, percentage, `auto`, or `anchor-size()`.		CSS Logical Properties 1
`margin-left`	Sets the space outside the left side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-right`	Sets the space outside the right side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-top`	Sets the space outside the top side of a box.	Length, percentage, `auto`, or `anchor-size()`.		CSS Box Model 4
`margin-trim`	Removes selected margins where child boxes meet the edges of their container.	none; `block`, `inline`, or both; or one or more of `block-start`, `inline-start`, `block-end`, and `inline-end`.		CSS Box Model 4
`padding`	Sets the space between the content and all four sides of its border.	One to four non-negative lengths or percentages.		CSS Box Model 4
`padding-block`	Sets the padding at the start and end of the block axis.	One or two non-negative lengths or percentages.		CSS Logical Properties 1
`padding-block-end`	Sets the padding at the end of the block axis.	Non-negative length or percentage.		CSS Logical Properties 1

Name	Description	Values	Default	Specification
`padding-block-start`	Sets the padding at the start of the block axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-bottom`	Sets the space between the content and the bottom border.	Non-negative length or percentage.		CSS Box Model 4
`padding-inline`	Sets the padding at the start and end of the inline axis.	One or two non-negative lengths or percentages.		CSS Logical Properties 1
`padding-inline-end`	Sets the padding at the end of the inline axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-inline-start`	Sets the padding at the start of the inline axis.	Non-negative length or percentage.		CSS Logical Properties 1
`padding-left`	Sets the space between the content and the left border.	Non-negative length or percentage.		CSS Box Model 4
`padding-right`	Sets the space between the content and the right border.	Non-negative length or percentage.		CSS Box Model 4
`padding-top`	Sets the space between the content and the top border.	Non-negative length or percentage.		CSS Box Model 4

Dimension properties

This reference lists the CSS sizing properties documented in Box dimensions. The `Values` column gives CSS syntax. CSSMedia accepts these declarations for computed styles, but it has no element layout stage and does not generate a PDF. The properties therefore have no PDF effect. The `Default` cells are left blank because there is no PDF output.

CSS Anchor Positioning permits `anchor-size()` only on properties accepted in an `@position-try` rule. These include sizing, inset, and margin properties. This page lists the sizing properties.

Preferred sizes

Name	Description	Values	Default	Specification
`width`	Sets the preferred physical width.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`height`	Sets the preferred physical height.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`inline-size`	Sets the preferred size on the inline axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`block-size`	Sets the preferred size on the block axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

- `<length-percentage>` means a length such as `12pt`, a percentage such as `60%`, or a `calc()` expression combining them.
- `[0,∞]` in `fit-content(<length-percentage [0,∞]>)` means the length or percentage cannot be negative.
- `anchor-size()` accepts an optional anchor name, an optional size keyword, or both, followed by an optional length or percentage fallback. It resolves to the anchor's size only when the box is absolutely positioned and has a target anchor. Otherwise, it resolves to the fallback. Without a fallback, the declaration becomes invalid at computed-value time. The sizing properties listed here include physical and logical properties.

Minimum and maximum sizes

Name	Description	Values	Default	Specification
`min-width`	Sets a lower limit for the physical width.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-height`	Sets a lower limit for the physical height.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-inline-size`	Sets a lower limit on the inline axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`min-block-size`	Sets a lower limit on the block axis.	`auto`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-width`	Sets an upper limit for the physical width.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-height`	Sets an upper limit for the physical height.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content(<length-percentage [0,∞]>)`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3
`max-inline-size`	Sets an upper limit on the inline axis.	`none`, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

Name	Description	Values	Default	Specification
`max-block-size`	Sets an upper limit on the block axis.	none, `<length-percentage>`, `stretch`, `contain`, `min-content`, `max-content`, `fit-content`, `fit-content()`, `calc-size()`, or `anchor-size()`.		CSS Sizing 3

- `<length-percentage>` means a length such as `12pt`, a percentage such as `60%`, or a `calc()` expression combining them.
- `[0,∞]` in `fit-content(<length-percentage [0,∞]>)` means the length or percentage cannot be negative.

Paired size shorthands

Name	Description	Values	Default	Specification
`size`	Sets `width` and `height`.	One or two preferred-size values.		CSS Sizing 4
`min-size`	Sets `min-width` and `min-height`.	One or two minimum-size values.		CSS Sizing 4
`max-size`	Sets `max-width` and `max-height`.	One or two maximum-size values.		CSS Sizing 4

- Each pair takes the corresponding longhand's values in width-then-height order. The second value is optional and defaults to the first.
- The `size` shorthand is not valid in an @page context. That context has a separate `size` descriptor for page dimensions.

Aspect ratio and intrinsic sizes

Name	Description	Values	Default	Specification
`aspect-ratio`	Supplies a preferred ratio between the width and height.	`auto`, a non-negative number such as `1.5`, or two non-negative numbers such as `16 / 9`; `auto` can be combined with a number.		CSS Sizing 4
`contain-intrinsic-width`	Fallback intrinsic width under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-height`	Fallback intrinsic height under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4

Name	Description	Values	Default	Specification
`contain-intrinsic-inline-size`	Fallback intrinsic inline size under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-block-size`	Fallback intrinsic block size under size containment.	Optional `auto`; none or a non-negative length.		CSS Sizing 4
`contain-intrinsic-size`	Sets fallback intrinsic width and height.	One or two longhand values; see below.		CSS Sizing 4

- For `aspect-ratio`, the first number is the width and the second is the height. If you give one number, the second number is `1`. For example, `1.5` means the same ratio as `1.5 / 1`.
- Each `contain-intrinsic-*` longhand accepts an optional `auto`, followed by none or a non-negative length.
- The `contain-intrinsic-size` shorthand takes one or two such values. One value applies to both dimensions.

Replaced elements and intrinsic contributions

Name	Description	Values	Default	Specification
`frame-sizing`	Selects a replaced element's content size for sizing.	`auto` or a `content-*` dimension.		CSS Sizing 4
`min-intrinsic-sizing`	Controls min-content contribution compression.	`legacy`, `zero-if-scroll`, `zero-if-extrinsic`, or both.		CSS Sizing 4

- `frame-sizing` accepts `auto`, `content-width`, `content-height`, `content-block-size`, or `content-inline-size`.

Background and image reference

Name	Description	Values	Default	Specification
`background`	Sets the background color and one or more background image layers. The color, if present, belongs to the final layer.	Comma-separated layers with image, position and optional size, repeat, attachment, and clipping or positioning box values; optional color on the final layer	See individual properties	CSS Backgrounds 4, `background`
`background-attachment`	Selects whether a background image is attached to its box, its contents, or the page box in paged media.	`scroll`, `fixed`, or `local`; comma-separated list	`fixed`	CSS Backgrounds 4, `background-attachment`
`background-blend-mode`	Sets how each background image layer blends with the layers below it and the background color.	`normal`, `darken`, `multiply`, `color-burn`, `lighten`, `screen`, `color-dodge`, `overlay`, `soft-light`, `hard-light`, `difference`, `exclusion`, hue, `saturation`, `color`, or `luminosity`; comma-separated list	`normal`	Compositing and Blending 1, `background-blend-mode`
`background-clip`	Sets the box area where the background is painted.	`border-box`, `padding-box`, `content-box`, `border-area`, `text`, or both `border-area` and `text`; comma-separated list	`border-box`	CSS Backgrounds 4, `background-clip`
`background-color`	Sets the background color.	`<color>`	`transparent`	CSS Backgrounds 4, `background-color`

Name	Description	Values	Default	Specification
`background-image`	Sets one or more background images or gradients.	One or more comma-separated none or `<image>` values	none	CSS Backgrounds 4, `background-image`
`background-origin`	Sets the box used as the positioning area for each background image.	`border-box`, `padding-box`, or `content-box`; comma-separated list	`padding-box`	CSS Backgrounds 4, `background-origin`
`background-position`	Sets the position of each background image.	One- to four-part position using keywords, lengths, or percentages; comma-separated list	`0% 0%`	CSS Backgrounds 4, `background-position`
`background-position-block`	Sets the position of each background image on the block axis.	`center`, or `start` or end, optionally combined with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-block`
`background-position-inline`	Sets the position of each background image on the inline axis.	`center`, or `start` or end, optionally combined with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-inline`
`background-position-x`	Sets the position of each background image on the physical horizontal axis.	`center`; `left`, `right`, `x-start`, or `x-end`; a length or percentage; or a side keyword with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-x`

Name	Description	Values	Default	Specification
`background-position-y`	Sets the position of each background image on the physical vertical axis.	`center`; `top`, `bottom`, `y-start`, or `y-end`; a length or percentage; or a side keyword with a length or percentage; comma-separated list	`0%`	CSS Backgrounds 4, `background-position-y`
`background-repeat`	Sets how each background image repeats along both axes.	`repeat`, `space`, `round`, or `no-repeat`; optional second value; `repeat-x`, `repeat-y`, `repeat-block`, or `repeat-inline`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat`
`background-repeat-block`	Sets how each background image repeats on the block axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-block`
`background-repeat-inline`	Sets how each background image repeats on the inline axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-inline`
`background-repeat-x`	Sets how each background image repeats on the physical horizontal axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-x`
`background-repeat-y`	Sets how each background image repeats on the physical vertical axis.	`repeat`, `space`, `round`, or `no-repeat`; comma-separated list	`repeat`	CSS Backgrounds 4, `background-repeat-y`

Name	Description	Values	Default	Specification
`background-size`	Sets the size of each background image.	One or two values, each `auto` or a non-negative length or percentage; or `cover` or `contain`; comma-separated list	`auto`	CSS Backgrounds 4, `background-size`
`background-tbd`	Sets background layer properties without setting the background color.	One or more background layers	See individual properties	CSS Backgrounds 4, `background-tbd`
`isolation`	Controls whether an element forms an isolated group for blending.	`auto` or `isolate`	`auto`	Compositing and Blending 1, `isolation`
`mix-blend-mode`	Sets how an element blends with the content behind it.	`normal`, `darken`, `multiply`, `color-burn`, `lighten`, `screen`, `color-dodge`, `overlay`, `soft-light`, `hard-light`, `difference`, `exclusion`, hue, `saturation`, `color`, or `luminosity`	`normal`	Compositing and Blending 1, `mix-blend-mode`

- `<color>` means a CSS color, such as navy, `#336699`, or `rgb(51 102 153)`.
- `<image>` means none, an image such as `url("paper.png")`, or a gradient such as `linear-gradient(white, gray)`.

Display and positioning

The `display` property determines the kind of box an element generates. Positioning properties describe how a box is placed relative to its normal position or containing block.

Name	Description	Values	Default	Specification
`display`	Selects whether an element generates a box and describes its outer and inner display types.	`none`, `contents`, or combinations of outer types (`block`, `inline`, `run-in`), inner types (`flow`, `flow-root`, `table`, `flex`, `grid`, `ruby`), and `list-item`; also internal table and ruby types, legacy combinations such as `inline-block`, `inline-flex`, and `inline-grid`, plus `math`, `grid-lanes`, and `inline-grid-lanes`		CSS Display 4 §2
`visibility`	Controls whether a box is visible while preserving its place in layout.	`visible`, `hidden`, `force-hidden`, or `collapse`		CSS Display 4 §5
`order`	Changes the order of flex and grid items for layout and painting.	An integer		CSS Display 4 §3
`reading-flow`	Chooses the reading order of children in a block, flex, or grid container.	`normal`, `source-order`, `flex-visual`, `flex-flow`, `grid-rows`, `grid-columns`, or `grid-order`		CSS Display 4 §4
`reading-order`	Assigns an integer ordering value to a direct child of a reading-flow container.	An integer		CSS Display 4 §4.1
`position`	Chooses the positioning scheme for an element.	`static`, `relative`, `absolute`, `sticky`, `fixed`, or `running( )`		CSS Position 3 §2; CSS GCPM 3 §1.2.1

Name	Description	Values	Default	Specification
`top`, `right`, `bottom`, `left`	Set an inset from the corresponding side of the containing block for a positioned element.	`auto`, a length or percentage, `anchor()`, or `anchor-size()`		CSS Position 3 §3.1
`inset`	Sets the four physical insets.	One to four values accepted by `top`, `right`, `bottom`, and `left`		CSS Position 3 §3.2
`inset-block`, `inset-inline`	Set the start and end insets along the block or inline axis.	One or two values accepted by the corresponding logical inset properties		CSS Position 3 §3.2
`inset-block-start`, `inset-block-end`, `inset-inline-start`, `inset-inline-end`	Set one inset on the block or inline axis.	`auto`, a length or percentage, `anchor()`, or `anchor-size()`		CSS Position 3 §3.1
`z-index`	Sets a positioned box's stack level.	`auto` or an integer		CSS 2 §9.9

- A length is a size with a unit, such as `12pt` or `1em`. A percentage is relative to the containing block, as defined for the relevant inset.
- `anchor()` uses an anchor's side or center as an inset coordinate. `anchor-size()` uses a selected anchor's size. See CSS Anchor Positioning 1 and the `anchor-size()` function.
- `running()` names an element to move into a page margin box. See CSS GCPM 3.
- An integer is a whole number without a fractional part. Negative values, zero, and positive values are allowed for `order` and `z-index`; CSS Display defines how ordering is applied.

Page reference

Name	Description	Values	Default	Specification
`page`	Assigns a named page type to a box.	`auto` or a custom identifier	`auto`	CSS Paged Media 3, page
`@page`	Defines page-box declarations and may contain page-margin rules.	Optional comma-separated page selectors		CSS Paged Media 3, @page

Name	Description	Values	Default	Specification
`size`	Sets page-box size and orientation. `auto` uses A4.	`auto`; one or two lengths; or A3, A4, A5, B4, B5, `JIS-B4`, `JIS-B5`, `ledger`, `legal`, or `letter`, optionally combined with `portrait` or `landscape`	A4	CSS Paged Media 3, `size`
`bleed`	Sets the bleed extension beyond the page box. `auto` uses `6pt` with crop marks and `0` otherwise.	`auto` or a length	`0` (`auto`)	CSS Paged Media 3, `bleed`
`marks`	Requests crop marks, registration marks, or both outside the page box.	none, `crop`, `cross`, or both `crop` and `cross`	none	CSS Paged Media 3, `marks`
`page-orientation`	Rotates the laid-out page in the output.	`upright`, `rotate-left`, or `rotate-right`	`upright`	CSS Paged Media 3, `page-orientation`
`@top-left-corner`	Defines the fixed-size box at the intersection of the top and left page margins.			CSS Paged Media 3, `@top-left-corner`
`@top-left`	Defines the box between the top-left corner and top-center boxes.			CSS Paged Media 3, `@top-left`
`@top-center`	Defines the centered box in the top page margin.			CSS Paged Media 3, `@top-center`
`@top-right`	Defines the box between the top-center and top-right corner boxes.			CSS Paged Media 3, `@top-right`

Name	Description	Values	Default	Specification
`@top-right-corner`	Defines the fixed-size box at the intersection of the top and right page margins.			CSS Paged Media 3, `@top-right-corner`
`@right-top`	Defines the box between the top-right corner and right-middle boxes.			CSS Paged Media 3, `@right-top`
`@right-middle`	Defines the centered box in the right page margin.			CSS Paged Media 3, `@right-middle`
`@right-bottom`	Defines the box between the right-middle and bottom-right corner boxes.			CSS Paged Media 3, `@right-bottom`
`@bottom-right-corner`	Defines the fixed-size box at the intersection of the bottom and right page margins.			CSS Paged Media 3, `@bottom-right-corner`
`@bottom-right`	Defines the box between the bottom-center and bottom-right corner boxes.			CSS Paged Media 3, `@bottom-right`
`@bottom-center`	Defines the centered box in the bottom page margin.			CSS Paged Media 3, `@bottom-center`
`@bottom-left`	Defines the box between the bottom-left corner and bottom-center boxes.			CSS Paged Media 3, `@bottom-left`
`@bottom-left-corner`	Defines the fixed-size box at the intersection of the bottom and left page margins.			CSS Paged Media 3, `@bottom-left-corner`

Name	Description	Values	Default	Specification
`@left-bottom`	Defines the box between the left-middle and bottom-left corner boxes.			CSS Paged Media 3, `@left-bottom`
`@left-middle`	Defines the centered box in the left page margin.			CSS Paged Media 3, `@left-middle`
`@left-top`	Defines the box between the top-left corner and left-middle boxes.			CSS Paged Media 3, `@left-top`

Declarations in page-margin rules

Each page-margin at-rule contains declarations written as `property: value;`. Use properties that apply to page-margin boxes, including `content`, font and text properties, color, backgrounds, borders, margins, padding, dimensions and `vertical-align`. See the CSS Paged Media page-margin property list.

For example, set a running title in the top-center box and a page number in the bottom-center box:

```
@page {
    @top-center {
        content: "Annual report";
        font-weight: bold;
    }

    @bottom-center {
        content: "Page " counter(page);
        text-align: center;
    }
}
```

Links and generated content reference

The values list the syntax defined by the cited specifications.

Name	Description	Values	Default	Specification
`content`	Replaces an element's contents or supplies generated content for pseudo-elements and page margin boxes.	`normal`, none, an image that replaces the contents, a list of generated items, or `element()`; a list can include alternative text after `/`		CSS Content 3 §1
`bookmark-label`	Sets the text for a generated bookmark.	A list of generated content items		CSS Content 3 §3.2
`bookmark-level`	Sets a bookmark's hierarchy level, or disables its creation.	none or an integer of 1 or greater		CSS Content 3 §3.1
`bookmark-state`	Sets whether a generated bookmark starts open or closed.	`open` or `closed`		CSS Content 3 §3.3
`quotes`	Selects the quotation marks used by generated `open-quote` and `close-quote` content.	`auto`, none, `match-parent`, or one or more pairs of quoted strings		CSS Content 3 §2.4.1
`string-set`	Assigns named values from an element for retrieval in page content.	none or comma-separated pairs of a name and one or more strings		CSS Content 3 §2.7.1
`link`	Sets the destination of an XML element as a CSSMedia extension.	none or one or more quoted strings or `url()` and `src()` values		CSSMedia extension; see Links
`link-parameters`	Supplies parameters for links to external resources.	none or a comma-separated list of `param()` values		CSS Link Parameters 1 §2.1

- A `<content-list>` is one or more generated content items, such as strings, quote keywords, counters, images, attribute values, or `content()` values. See CSS Content 3 and Named strings.
- A `<content-replacement>` is one image that replaces the element's normal contents. See CSS Content 3.
- The image form of `element()` uses an element ID to produce an image; the running-element form selects a named element for use in page margin boxes. See CSS Images 4 and CSS GCPM 3.
- A positive `<integer>` for `bookmark-level` is a whole number of at least 1. The top level is 1.

- A `<string>` is text between quotes. `url()` and `src()` provide URL values.
- A `<custom-ident>` is an author-chosen identifier, not a CSS-wide keyword or none.
- `param()` takes a dashed parameter name beginning with `--` and an optional `<declaration-value>`. If omitted, the parameter value is empty. A `<declaration-value>` is the sequence of CSS component values after a declaration's colon, excluding top-level semicolons and unmatched closing braces. See CSS Link Parameters 1 and CSS Syntax 3.

For examples of destinations, named strings, and generated content, see Links and Named strings.

Counters, list markers, and footnotes

Name	Description	Values	Default	Specification
`counter-reset`	Creates a new counter instance and initializes it.	none or one or more counter names, each optionally followed by an integer; the default value for each name is `0`		CSS Lists 3 §4.1
`counter-increment`	Increases each named counter, creating it if needed.	none or one or more counter names, each optionally followed by an integer; the default increment is `1`		CSS Lists 3 §4.2
`counter-set`	Sets the value of existing counters, creating a counter if it does not exist.	none or one or more counter names, each optionally followed by an integer; the default value is `0`		CSS Lists 3 §4.3
`list-style`	Sets the marker's position, image, and type.	Any order of `inside` or `outside`, an image or none, and a counter style, string, or none; any component can be omitted		CSS Lists 3 §3.6
`list-style-image`	Uses an image as the list marker.	An image or none		CSS Lists 3 §3.1
`list-style-position`	Places the marker inside the list item's contents or outside its principal box.	`inside` or `outside`		CSS Lists 3 §3.5

Name	Description	Values	Default	Specification
`list-style-type`	Chooses the marker's counter style or a literal marker string.	A counter-style name, a string, or none		CSS Lists 3 §3.4
`marker-side`	Chooses whether an outside marker's position follows the list item's directionality or its parent's.	`match-self` or `match-parent`		CSS Lists 3 §3.7
`@counter-style`	Defines a named counter style for markers and counter functions.	A name followed by counter-style descriptors		CSS Counter Styles 3 §3
`system` descriptor	Selects the algorithm that converts counter values to symbols.	`cyclic`, `numeric`, `alphabetic`, `symbolic`, `additive`, `fixed` with an optional starting integer, or `extends` with another counter-style name		CSS Counter Styles 3 §3.1
`symbols` descriptor	Lists the symbols used by the cyclic, fixed, symbolic, alphabetic, or numeric systems.	One or more symbols		CSS Counter Styles 3 §3.2
`additive-symbols` descriptor	Maps descending counter weights to symbols for the additive system.	Comma-separated pairs of a non-negative integer and a symbol, in descending weight order		CSS Counter Styles 3 §3.3
`negative` descriptor	Adds symbols before and optionally after a negative counter representation.	One or two symbols		CSS Counter Styles 3 §3.4
`prefix` descriptor	Adds a symbol before each marker.	One symbol		CSS Counter Styles 3 §3.7
`suffix` descriptor	Adds a symbol after each marker.	One symbol		CSS Counter Styles 3 §3.7
`range` descriptor	Limits the values that the counter style can represent before fallback is used.	`auto` or comma-separated pairs of integer bounds or `infinite`		CSS Counter Styles 3 §3.5

Name	Description	Values	Default	Specification
`pad` descriptor	Pads short representations to a minimum length with a chosen symbol.	A non-negative integer and a symbol		CSS Counter Styles 3 §3.6
`fallback` descriptor	Names the counter style used when this style cannot represent a value.	A counter-style name		CSS Counter Styles 3 §3.8
`footnote-display`	Chooses the layout form used to display a footnote.	`block`, `inline`, or `compact`		CSS GCPM 3 §2.6
`footnote-policy`	Controls how footnote content is kept together when it does not fit in the remaining page space.	`auto`, `line`, or `block`		CSS GCPM 3 §2.8

- A <`counter-name`> is an identifier naming a CSS counter. none and CSS-wide keywords are not counter names. See CSS Lists 3.
- An <`integer`> is a whole number without a fractional part. In the `range` descriptor, `infinite` means there is no bound in that direction. See CSS Values 4.
- An <`image`> is a URL, image function, or gradient that can provide image data. See CSS Images 3.
- A <`counter-style`> is a predefined counter style name, a name defined by `@counter-style`, or a `symbols()` function. See CSS Counter Styles 3.
- A <`counter-style-name`> is the identifier naming a predefined or author-defined counter style. See CSS Counter Styles 3.
- A <`symbol`> is a string, an identifier whose characters form the symbol, or an image. See CSS Counter Styles 3.

For example, this rule defines a two-symbol cycle. Replace `item` with an element name in your XML. The `display` declaration makes those elements list items, and their markers use the defined style.

```
@counter-style stars {
    system: cyclic;
    symbols: "★" "□";
    suffix: " ";
}

item {
    display: list-item;
    list-style-type: stars;
}
```

CSSMedia applies counter operations and prepares text marker content while building the box tree.

Cascade reference

The `Default` column lists values used by CSSMedia's current PDF-oriented rule evaluation. Media-query width, height and ratio values use the A4 default page size.

Property and at-rule reference

Name	Description	Values	Default	Specification
`all`	Resets all properties except `direction` and `unicode-bidi`.	`initial` | `inherit` | `unset` | `revert` | `revert-layer`		CSS Cascade 5 §3.2
`@charset`	Declares the stylesheet's fallback character encoding.	`@charset "<label>";`		CSS Syntax 3 §8.3
`@import`	Loads another stylesheet, optionally conditioned on a layer, support test or media query.	`@import <url-or-string> [layer | layer(<layer-name>)]? [supports(<condition>)]? <media-query-list>?;`		CSS Cascade 5 §2
`@namespace`	Declares a namespace prefix for selectors.	`@namespace <namespace-prefix>? <string-or-url>;`		CSS Namespaces 3 §3
`@layer`	Declares layer order or groups rules in a cascade layer.	`@layer <layer-name>? { <stylesheet> }` or `@layer <layer-name-list>;`		CSS Cascade 5 §6.4

Name	Description	Values	Default	Specification
`@property`	Registers a custom property with syntax, inheritance and optional initial value.	`@property <custom-property-name> { <declaration-list> }`		CSS Properties and Values API 1 §3
`syntax`	Defines the grammar of a registered custom property.	`<string>`		CSS Properties and Values API 1 §3.1
`inherits`	Sets whether a registered custom property inherits by default.	`true` | `false`		CSS Properties and Values API 1 §3.2
`initial-value`	Sets the registered custom property's initial value.	`<declaration-value>`	Guaranteed-invalid for `syntax: "*"`; otherwise required	CSS Properties and Values API 1 §3.3
`@media`	Includes rules when a media query matches the PDF environment.	`@media <media-query-list> { <rule-list> }`		CSS Conditional 3 §5
`@supports`	Includes rules when CSSMedia accepts a declaration, selector or supported feature test.	`@supports <supports-condition> { <rule-list> }`		CSS Conditional 3 §6
`@supports-condition`	Defines a named group of support tests.	`@supports-condition <supports-condition-name> { <block-contents> }`		CSS Conditional 5 §8

Name	Description	Values	Default	Specification
`@container`	Includes rules when a selected ancestor container passes the query.	`@container <container-condition># { <block-contents> }`		CSS Conditional 5 §5.4
`@scope`	Limits the selectors in a rule list to a scope root and optional scope limit.	`@scope [(<scope-start>)]? [to (<scope-end>)]? { <rule-list> }`		CSS Cascade 6 §2.5
`@custom-media`	Gives a name to a media query list or boolean value.	`@custom-media <extension-name> [<media-query-list> | true | false];`		Media Queries 5 §10
`@when`	Includes rules when a generalized boolean condition is true.	`@when <boolean-condition> { <rule-list> }`		CSS Conditional 5 §3
`@else`	Supplies the next branch of a conditional chain.	`@else <boolean-condition>? { <rule-list> }`		CSS Conditional 5 §4
`@function`	Defines a custom CSS function.	`@function <name>(<parameters>) [returns <type>]? { <declaration-rule-list> }`		CSS Functions and Mixins 1 §2.1
`result`	Supplies a custom function's return value.	`<declaration-value>?`		CSS Functions and Mixins 1 §2.2

- ? means optional, | means choose one, # means one or more comma-separated items, || means one or more options in any order, and brackets group parts of the syntax.
- `<label>` is an encoding name.
- `<url-or-string>` and `<string-or-url>` each mean a URL or quoted string.
- `<namespace-prefix>` is the prefix bound to a namespace.

- `<layer-name>` names a cascade layer. `<layer-name-list>` is a comma-separated list of layer names.
- `<stylesheet>` is a CSS stylesheet.
- `<custom-property-name>` is a custom property name beginning with `--`.
- `<declaration-list>` is a list of declarations.
- `<block-contents>` is content inside a rule's braces, including declarations and nested rules.
- `<declaration-value>` is the value part of a declaration: one or more CSS tokens, such as `navy`, `calc(1em + 2px)` or `var(--size)`. It cannot contain a top-level semicolon or `!`, since those delimit declarations and `!important`. Semicolons and exclamation marks inside functions are not top-level. See CSS Syntax 3.
- `<string>` is a quoted CSS string.
- `<media-query-list>` is one or more comma-separated media queries.
- `<rule-list>` is a sequence of rules.
- `<condition>` and `<supports-condition>` are support conditions made from feature tests and boolean operators.
- `<supports-condition-name>` names a reusable support condition.
- `<container-condition>` optionally names a query container and gives its query.
- `<boolean-condition>` combines `media()` and `supports()` tests.
- `<scope-start>` and `<scope-end>` are selector lists that set the scope roots and limits of `@scope`.
- `<extension-name>` names a custom media query.
- `<name>` in the function syntax is a function name beginning with `--`.
- `<parameters>` is a comma-separated list of function parameters.
- `<type>` is the optional syntax expected for the function result.
- `<declaration-rule-list>` contains declarations and nested rules.
- `<number>` in the function example is a CSS number, which can be an integer or include a fractional part.

The `@charset` rule must be the first rule in a stylesheet to affect byte decoding. Use `@namespace` before ordinary rules, and place `@import` rules before namespace declarations and ordinary rules.

Function names start with `--`. Parameters can have a syntax type and a default value. Use the `result` descriptor for the returned value:

```
@function --double(--value <number>) returns <number> {
    result: calc(var(--value) * 2);
}
```

`@supports` declaration tests use parentheses. `selector()` tests one complex selector. `font-tech()`, `font-format()` and `at-rule()` test supported features. A named condition is tested by placing its name inside parentheses:

```
@supports (color: navy) and selector(chapter > title) {
    chapter > title { color: navy; }
}

@supports-condition --has-navy {
    color: navy;
}

@supports (--has-navy) {
    chapter { color: navy; }
}
```

`@when` conditions combine `media()` and `supports()` tests with `and`, `or` and `not`. Parentheses group mixed operators:

```
@when media(width >= 6in) and supports(color: navy) {
    chapter { color: navy; }
}
```

Use `@else` immediately after an `@when`, `@media`, `@supports` or `@container` branch. An unrelated rule ends the chain.

Media features

Media feature values describe the page or PDF environment used for conditional rule evaluation. They do not describe a screen.

Name	Description	Values	Default	Specification
`any-hover`	Whether any pointing device can hover.	`none` | `hover`	`none`	Media Queries 5
`any-pointer`	The accuracy of any available pointing device.	`none` | `coarse` | `fine`	`none`	Media Queries 5
`aspect-ratio`	The ratio of page width to page height.	`<ratio>`	A4 page ratio	Media Queries 5
`color`	Bits per color component in the output.	`<integer>`	`8`	Media Queries 5
`color-gamut`	The approximate output color gamut.	`srgb` | `p3` | `rec2020`	`srgb`	Media Queries 5
`color-index`	Number of entries in the output color lookup table.	`<integer>`	`0`	Media Queries 5

Name	Description	Values	Default	Specification
`device-aspect-ratio`	The ratio of output width to output height.	`<ratio>`	A4 page ratio	Media Queries 5
`device-height`	Output height.	`<length>`	A4 page height	Media Queries 5
`device-width`	Output width.	`<length>`	A4 page width	Media Queries 5
`display-mode`	The presentation mode.	`fullscreen` | `standalone` | `minimal-ui` | `browser` | `picture-in-picture`	Not set	Media Queries 5
`dynamic-range`	The output's dynamic range.	`standard` | `high`	`standard`	Media Queries 5
`environment-blending`	Whether output blends with the surrounding environment.	`opaque` | `additive` | `subtractive`	`opaque`	Media Queries 5
`forced-colors`	Whether the output forces a limited color palette.	`none` | `active`	`none`	Media Queries 5
`grid`	Whether the output is grid-based.	`<mq-boolean>` (`0` or `1`)	`0`	Media Queries 5
`height`	Page height.	`<length>`	A4 page height	Media Queries 5
`horizontal-viewport-segments`	Number of horizontal viewport segments.	`<integer>`	`1`	Media Queries 5
`hover`	Whether the primary pointing device can hover.	`none` | `hover`	`none`	Media Queries 5
`inverted-colors`	Whether the output inverts colors.	`none` | `inverted`	`none`	Media Queries 5

Name	Description	Values	Default	Specification
`monochrome`	Number of bits per pixel in a monochrome output.	`<integer>`	`0`	Media Queries 5
`nav-controls`	Availability of navigation controls.	`none` | `back`	`none`	Media Queries 5
`orientation`	Whether the page is portrait or landscape.	`portrait` | `landscape`	`portrait`	Media Queries 5
`overflow-block`	How the output handles block-axis overflow.	`none` | `scroll` | `paged`	`paged`	Media Queries 5
`overflow-inline`	How the output handles inline-axis overflow.	`none` | `scroll`	`none`	Media Queries 5
`pointer`	The accuracy of the primary pointing device.	`none` | `coarse` | `fine`	`none`	Media Queries 5
`prefers-color-scheme`	The preferred color scheme.	`light` | `dark`	`light`	Media Queries 5
`prefers-contrast`	The preferred contrast level.	`no-preference` | `less` | `more` | `custom`	`no-preference`	Media Queries 5
`prefers-reduced-data`	Whether the user prefers reduced data use.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`prefers-reduced-motion`	Whether the user prefers reduced motion.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`prefers-reduced-transparency`	Whether the user prefers reduced transparency.	`no-preference` | `reduce`	`no-preference`	Media Queries 5
`resolution`	Output pixel density.	`<resolution>` | `infinite`	`infinite`	Media Queries 5
`scan`	The output's scanning process.	`interlace` | `progressive`	Not set	Media Queries 5

Name	Description	Values	Default	Specification
`scripting`	Whether scripting is available.	`none` | `initial-only` | `enabled`	`none`	Media Queries 5
`update`	How frequently the output can update.	`none` | `slow` | `fast`	`none`	Media Queries 5
`vertical-viewport-segments`	Number of vertical viewport segments.	`<integer>`	`1`	Media Queries 5
`video-color-gamut`	The color gamut of the output video.	`srgb` | `p3` | `rec2020`	`srgb`	Media Queries 5
`video-dynamic-range`	The dynamic range of the output video.	`standard` | `high`	`standard`	Media Queries 5
`width`	Page width.	`<length>`	A4 page width	Media Queries 5

- `<ratio>` is a ratio of two numbers.
- `<integer>` is a whole number.
- `<length>` is a CSS length, such as `12pt` or `2em`.
- `<mq-boolean>` is an integer with the value `0` or `1`.
- `<resolution>` is a pixel density, such as `300dpi`.

Container style queries

A container style query uses `style()` to test a computed style or custom property on an ancestor. See Conditional rules.

Licenses

CSSMedia

- Available from https://github.com/maxprograms-com/CSSMedia
- Published under Business Source License 1.1

Adobe Core 14 AFM files

- Available from https://download.macromedia.com/pub/developer/opentype/tech-notes/Core14_AFMs.zip
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Adobe Glyph List and ITC Zapf Dingbats Glyph List

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Unicode Character Database

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- Available from https://github.com/unicode-org/cldr-json

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Universal Shaping Engine data

- Available from https://github.com/microsoft/font-tools/tree/main/USE
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W3C CSS Counter Styles Level 3

- Available from https://www.w3.org/TR/css-counter-styles-3/
- Published under W3C Software and Document License

W3C webref

- Available from https://github.com/w3c/webref
- Published under MIT License

Glossary

CSS

Cascading Style Sheets (CSS) is the language, defined by the W3C, that describes how elements of a document are presented. A stylesheet is a list of rules. Each rule has a selector that chooses elements and declarations that set their properties. CSSMedia reads a CSS file to decide how to format an XML document in a PDF.

DTD

A Document Type Definition (DTD) is a grammar, defined by the XML specification, that declares the elements and attributes an XML document may use. A document points to its DTD in its document type declaration. When the DTD is not at the location the document names, an OASIS Catalog tells CSSMedia where to find it.

MathML

Mathematical Markup Language (MathML) is a W3C XML language for writing mathematical notation. A MathML formula describes the structure of an expression, such as fractions, roots and subscripts. CSSMedia draws MathML found in a document as an image.

OASIS Catalog

An OASIS Catalog is an XML file, defined by the OASIS XML Catalogs standard, that maps the identifiers used in documents to the locations where the files are stored. A catalog lets a parser find a DTD, an XML Schema or a RelaxNG schema without downloading it. Pass a catalog to CSSMedia with the `-g` option.

PDF

Portable Document Format (PDF) is the file format, standardized as ISO 32000, for documents that look the same on any device. A PDF holds pages with text, fonts, images and vector graphics at fixed positions, and may also hold links and bookmarks. CSSMedia writes a PDF file from an XML file and a CSS file.

RelaxNG

RelaxNG is a schema language for XML, standardized as ISO/IEC 19757-2. A RelaxNG schema describes the structure of a document in an XML syntax or in a compact non-XML syntax. It is an alternative to a DTD and to an XML Schema, and it can be located with an OASIS Catalog.

SVG

Scalable Vector Graphics (SVG) is a W3C format for two-dimensional vector graphics, written as XML. An SVG drawing describes shapes, paths, text and images that can be scaled without losing quality. CSSMedia draws SVG found in a document as an image.

XML

Extensible Markup Language (XML) is a W3C standard for writing structured documents. An XML document is a tree of elements, with attributes and text, whose names are chosen by the author or by a vocabulary such as DITA or DocBook. CSSMedia reads an XML file as the content of the PDF.

XML Schema

XML Schema is a W3C language for describing the structure of XML documents and the data types of their content. A schema is itself an XML file, and it is more expressive than a DTD. Like a DTD, a schema can be located with an OASIS Catalog.